Bank Market Power and Interest Rate Risk *

Yilun Allen Shen †

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Abstract

How does relative market power in funding and lending shape bank interest rate risk exposure and monetary transmission? We document a novel pricing fact: banks with less rate-sensitive funding charge higher fixed-rate loan spread over variable-rate loans, despite stable funding makes fixed-rate lending an attractive earnings hedge. Greater loan market concentration is associated with higher spreads conditional on funding sensitivity and a weaker expansion of fixed-rate credit after monetary tightening. We develop a two-date model in which banks trade higher expected earnings from restricting fixed-rate lending against greater earnings risk due to forgone hedging. A restriction on relative market power reproduces the negative premium–sensitivity slope, while endogenous entry explains why competition in both markets moves together. A calibrated investment extension captures lower premiums and a higher fixed-rate share after an interest rate increase. The extension implies a contraction in aggregate investment. An application shows that greater bank risk-bearing capacity can raise expected earnings and earnings volatility while reducing fixed-rate lending and investment. Banks that can bear more risk become less willing to supply borrower protection, linking bank risk management to monetary transmission.

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†University College London. Email: allen.shen.16@ucl.ac.uk.

1 Introduction

The banking turmoil of March 2023, following the rapid rise in U.S. interest rates, renewed concerns about banks' exposure to interest rate risk. Deposit market power dampens the passthrough of higher policy rates to funding costs and hence limits the increase in interest expense. However, the effect on bank earnings also depends on interest income responses, which reflect banks' portfolio choices and pricing incentives in loan markets. How does bank market power on both sides of the balance sheet jointly shape banks' interest rate risk exposure? And how does bank risk management affect monetary transmission?

We document a novel pricing fact: banks with less rate-sensitive funding charge higher fixed-rate premiums, defined as the difference between the origination rates on fixed- and variable-rate loans. A one-standard-deviation decrease in interest expense sensitivity is associated with a 5.3-basis-point higher fixed-rate premium. This pattern is surprising because stable funding costs make fixed-rate income an attractive earnings hedge. Taken alone, the hedging incentive therefore predicts *lower* fixed-rate premium at banks with insensitive funding, the opposite of the observed pricing relationship.

Empirical evidence points to the joint roles of funding and lending market power in this pricing relationship. We show that banks with more rate-insensitive funding operate in more concentrated local lending markets. Using U.S. bank and loan-level data, we find that a one-standard-deviation increase in bank lending market concentration is associated with a 22.5-basis-point higher premium, conditional on funding sensitivity. Empirical results also confirm bank hedging incentive: higher funding rate-sensitivity is associated with a higher premium, conditional on lending market concentration. These findings suggest that the fixed-rate premium reflects both funding and lending market power.

To interpret these facts, we build a two-date model in which risk-averse banks choose the composition of lending and borrowers select fixed- or variable-rate loans. Depositors search and compare offers locally, generating funding-rate passthrough that increases with deposit competition. Banks compete in quantities in a wider fixed-rate lending market, while variable-rate lending is competitive. The central mechanism is that *relative* market power in funding and lending determines banks' chosen exposure to interest rate risk. Funding market power governs interest expense sensitivity and hence the fixed-rate loan share that hedges earnings. Lending market power can make it profitable to restrict fixed-rate lending and sustain a larger spread over variable-rate loans. Banks thus trade higher expected earnings against forgone hedging and potential losses following adverse interest rate movements. When lending market power is sufficiently strong relative to funding market power, the model generates a negative premium–sensitivity slope, in line with our stylised fact. This mechanism complements existing theories of bank interest rate exposure (Di Tella & Kurlat, 2021; Hellwig, 1994) and extends the funding-based hedging motive of Drechsler et al. (2021) to

strategic loan pricing. The model also shows how lending market power impairs risk-sharing between banks and borrowers, with a welfare cost that is non-monotonic in interest rate volatility.

We then endogenise bank entry to explain why funding sensitivity and lending competition vary together. A larger market allows banks to spread a fixed participation cost over more business and supports more entrants. Entry makes deposit funding more rate-sensitive while increasing competition to supply fixed-rate protection. These changes have opposing effects on premiums: more responsive funding weakens the hedging incentive, while stronger lending competition reduces the rents from restricting fixed-rate credit. In the symmetric entry benchmark, market size selects the degree of competition along the premium–sensitivity relationship derived above. Entry therefore provides an economic source of variation in both markets, preserving the restriction on relative market power that determines the slope.

A second empirical fact concerns lending composition and monetary transmission. Using the monetary policy shocks of Jarociński and Karadi (2020), we find that fixed-rate loan stocks expand while variable-rate loan stocks initially contract following monetary tightening. Conditional on funding sensitivity, the fixed-rate lending response is weaker at banks with greater lending market power. Following a shock scaled to 100 basis points, the ten-quarter fixed-rate response is approximately 11% at low-HHI banks, compared with about 4.5% at high-HHI banks. These loan-stock responses combine lending, repayment and borrower-demand adjustments. Together with the pricing evidence, they motivate examining how market power interacts with bank risk management to shape monetary transmission.

To study the implications for real activity, we extend the model to endogenous project investment, household saving and labour, with heterogeneous local markets connected through interbank funding. Fixed-rate protection permits borrowers to undertake larger projects, linking loan pricing to investment. The effective cost of bank earnings risk rises with the expected financing rate, capturing a reduced-form deterioration in risk-bearing capacity. We calibrate three representative markets to funding sensitivities, loan premiums and fixed-rate shares. The calibration also targets a 3-basis-point decline in the mean premium and a 1.3-percentage-point increase in the fixed share of new loan volume after a 50-basis-point increase in the expected real rate. In this benchmark, fixed lending rises by 6.03%, while total investment falls by 0.67%. Cheaper protection supports fixed-financed projects, but does not offset the contraction from higher overall financing costs.

Finally, we apply the framework to differences in banks' capacity to bear risk. Greater capacity lowers the cost of retaining earnings exposure, weakening the incentive to supply fixed-rate loans as a hedge. Banks can then restrict protection and preserve higher premiums even when funding and lending competition are unchanged. In an experiment holding expected rates, uncertainty and the inherited terminal wage fixed, an 80% increase in effective capacity reduces fixed-rate lending by 26.01% and investment by approximately 0.02%.

Banking-sector expected earnings rise by 1.00%, while the standard deviation of earnings rises by 3.06%. Banks accept more earnings risk in exchange for a higher expected level of earnings. Greater dispersion in borrower risk aversion amplifies the lending and investment response locally.

Diversification and anticipated public support are possible sources of greater effective risk-bearing capacity. The application shows how banks that can bear more risk may provide less protection to borrowers, changing both the composition and quantity of credit. Assessing these changes requires considering lending terms and borrower investment alongside gains in bank resilience. Relative market power in funding and lending is central to this assessment because it determines how banks translate their capacity to bear risk into loan prices, portfolio choices and monetary transmission.

1.1 Related Literature

Our paper contributes to three strands of literature: theories of bank interest rate risk, bank market power, and risk sharing through loan contracts.

Theories of bank interest rate risk. Hellwig (1994) studies the efficient allocation of interest rate risk under liquidity provision, showing that eliminating exposure can sacrifice reinvestment opportunities. Drechsler et al. (2021) show that fixed-rate assets can stabilise earnings when deposit franchises insulate funding costs from market rates. Di Tella and Kurlat (2021) explain exposure through a dynamic hedging motive: higher rates raise deposit-intermediation profits, making banks willing to accept contemporaneous capital losses. Varraso (2025) links endogenous maturity choices to bank net worth and financial constraints in a competitive intermediation model.

We identify a complementary mechanism through loan-market pricing power. Banks trade the earnings hedge from fixed-rate lending against the rents preserved by restricting its supply. Relative funding and lending market power governs this tradeoff, linking banks' chosen exposure to the price of borrower protection. Our focus on earnings exposure differs from the market-value sensitivity of bank franchises emphasised by DeMarzo et al. (2024).

Bank market power. A large and growing literature studies bank market power on both sides of the balance sheet. Funding-market power shapes monetary transmission (Drechsler et al., 2017) and financial fragility (Egan et al., 2017), while lending-market power affects mortgage-rate pass-through and refinancing (Scharfstein & Sunderam, 2016), as well as corporate borrowing rates and aggregate activity (Bocola et al., 2026). Wang et al. (2022) jointly model deposit and loan competition to study monetary transmission. Hachem and Song (2021) show how interbank market power shapes liquidity choices and can make tighter liquidity regulation generate a credit boom.

We contribute by examining how funding and lending power jointly determine banks' interest rate risk management, motivated by our evidence that lending concentration is associated with higher fixed-rate premiums conditional on funding sensitivity. As in Choi and Rocheteau (2023), we ground deposit market power in search frictions. We combine deposit search with Cournot lending competition and endogenous entry, linking competition in both markets to banks' exposure and the supply of borrower protection.

Loan contracts and risk sharing. Fixed- and variable-rate loans allocate interest rate risk between banks and borrowers (Santomero, 1983). Vickery (2008) finds that financially constrained small firms are more likely to borrow at fixed rates, consistent with insurance demand. Kirti (2020) shows that banks with more rate-sensitive liabilities supply more floating-rate credit and price it more favourably relative to fixed-rate credit. Our evidence points to the opposite pricing relationship. Differences in lending competition may reconcile these findings: Kirti's pricing evidence uses residential mortgage quotes, whereas we study small business loans. Our model reproduces both relationships by varying relative funding and lending market power, highlighting its role in allocating interest rate risk.

Liu and Elenev (2025) connect mortgage fixation to financial stability and risk sharing; Abad et al. (2026) examine how loan fixation and bank solvency shape monetary transmission; and Bretscher et al. (2018) link interest rate uncertainty to corporate hedging and investment.

We extend these accounts of borrower insurance demand and banks' funding incentives by jointly determining fixed-rate loan pricing, borrower contract choice, and investment in our quantitative model. Greater bank risk-bearing capacity weakens the incentive to hedge earnings through fixed-rate lending. Banks restrict fixed credit and raise premiums, accepting greater earnings risk for higher expected earnings. More expensive protection reduces investment. This mechanism connects banks' capacity to bear risk to the supply of borrower protection and real activity.

2 Data and Empirical Facts

2.1 Data

Our empirical analysis and model calibration combine three main sources of data: loan-level records from the U.S. Small Business Administration's (SBA) 7(a) programme, bank Call Reports, and the Federal Deposit Insurance Corporation's (FDIC) Summary of Deposits. These sources provide information on the terms of individual business loans, banks' financial positions and interest flows, and the market structure of their branch networks. We focus on the sample period 2010Q1–2019Q4.

2.1.1 U.S. Small Business Administration 7(a) loans

We use loan-level administrative records from the SBA's 7(a) programme, which supports small businesses unable to obtain conventional credit on reasonable terms. Participating lenders originate loans, while the SBA guarantees a portion of each loan. Proceeds finance working capital, equipment, business premises and acquisitions, as well as refinancing existing debt. Loans may carry fixed or variable interest rates, negotiated between lenders and borrowers subject to programme limits. For variable-rate loans, the repricing frequency is also negotiated between borrower and lender, subject to the requirement that rates adjust no more frequently than monthly. For a detailed overview of the programme, its borrowers and lenders, and loan characteristics, see Gong et al. (2025).

We focus on loans approved during 2010Q1–2019Q4 under the main *Guaranty* subprogramme and issued by FDIC-insured depository institutions. The records identify the lender and report the initial interest rate, rate type, approved and guaranteed amounts, maturity, and borrower location, industry and jobs supported. Lender identifiers allow us to connect loan terms to banks' regulatory reports and branch networks. About one fifth of loans carry fixed rates, both by number and value. Appendix A.1 provides detailed summary statistics.

Although SBA lending accounts for a relatively small share of aggregate bank lending, these records provide a detailed window into small-business credit and the allocation of interest rate risk. The broader market for small loans is substantial: loans below $1 million represented approximately 22% of domestic banks' gross commercial and industrial credit extensions by value in the Federal Reserve's May 2017 survey.[1] Small firms' informational opacity, dependence on bank finance and limited use of derivatives make their loan terms particularly consequential for managing interest rate exposure (Vickery, 2008). These features make small-business lending a useful setting for studying lending market power.

2.1.2 Call Reports and FR Y-9C

We obtain quarterly balance-sheet and income-statement data from the FFIEC 031 Call Reports and the Federal Reserve's Consolidated Financial Statements for Holding Companies (FR Y-9C). Call Reports provide detailed financial information for individual reporting banks, including their assets, liabilities, capital, interest income and interest expense. The FR Y-9C provides corresponding information on a consolidated basis for reporting holding companies. These sources underpin our measures of banks' interest income and expense sensitivities and supply the balance-sheet characteristics used in the loan-pricing analysis.

We consolidate the data at the holding company level using institutional relationship information from the Federal Financial Institutions Examination Council (FFIEC). This unit of observation captures the scope for funding and asset allocation to be coordinated across

[1] Calculated from Table 2 of the Federal Reserve's *Survey of Terms of Business Lending*, May 1–5, 2017.

affiliated institutions. A holding company can raise funds on behalf of its subsidiaries, while internal capital markets allow resources to be reallocated within the group, influencing subsidiaries' lending decisions (Campello, 2002). Analysing subsidiaries in isolation could therefore omit relevant group-level funding costs and obscure the financial resources supporting their lending. Throughout the paper, "bank" refers to the consolidated entity.

We measure interest income and expense as annualised quarterly flows divided by quarterly average assets. Total interest expense includes payments on deposits, wholesale funding and other liabilities. The resulting expense sensitivity therefore captures the response of overall funding costs per dollar of assets. The regulatory reports also provide the bank size, capital and repricing information used as controls in the empirical analysis.

Following Drechsler et al. (2021), we estimate bank-specific interest expense sensitivities using a distributed-lag specification:

$$\Delta \text{IntExp}_{b,t} = \alpha_b + \eta_t + \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}} \Delta \text{FedFunds}_{t-\tau} + \varepsilon_{b,t}.$$

Here, $\text{IntExp}_{b,t}$ is bank b's annualised interest expense rate: total quarterly interest expense, including payments on deposits, wholesale funding and other liabilities, divided by quarterly average assets and multiplied by four. FedFunds_t denotes the effective federal funds rate, and Δ denotes a quarterly change. The terms α_b and η_t are bank and quarter fixed effects, respectively, and $\varepsilon_{b,t}$ is an error term. The bank-specific coefficients allow funding costs to respond differently across banks, while the contemporaneous rate change and three lags capture adjustment over four quarters.

We summarise each bank's interest expense sensitivity by the sum of these coefficients,

$$\beta_b^{\text{exp}} = \sum_{\tau=0}^{3} \beta_{b,\tau}^{\text{exp}}.$$

This measure captures the sensitivity of total funding expense per dollar of assets, rather than the passthrough to deposit rates alone. We construct interest income sensitivity, β_b^{inc}, analogously, replacing the dependent variable with the quarterly change in the annualised interest income rate, defined as four times total quarterly interest income divided by quarterly average assets.

2.1.3 Summary of Deposits (SOD)

The FDIC Summary of Deposits (SOD) reports annual branch-level information of FDIC-insured institutions as of June 30. It records branch deposit balance, location and institutional identifiers, allowing us to characterise banks' geographic networks and measure local banking market concentration. We treat counties as local markets and branches belonging

to the same holding company as a single competitor, consistent with Section 2.1.2.

For county c and year t, let $N_{b,c,t}$ denote bank b's number of branches in that county and let $N_{c,t} = \sum_b N_{b,c,t}$ denote the total number of bank branches in county c. We measure county-level concentration using the Herfindahl–Hirschman index (HHI),

$$\mathrm{HHI}_{c,t} = \sum_b \left(\frac{N_{b,c,t}}{N_{c,t}} \right)^2 .$$

We then construct a bank-level measure by averaging county HHIs across the bank's branch network, $\mathcal{C}(b)$, weighting each county equally:

$$\mathrm{HHI}_{b,t} = \frac{1}{|\mathcal{C}(b)|} \sum_{c \in \mathcal{C}(b)} \mathrm{HHI}_{c,t},$$

This measure summarises the concentration of banking markets across the bank's geographic network, which we use later as a proxy for bank lending market power, once conditional on bank funding sensitivity $\beta_b^{\exp}$.

We use $\mathrm{HHI}_{b,t}$ as a proxy for lending-market competition conditional on interest expense sensitivity, $\beta_b^{\exp}$. Local competition for borrowers can differ from the competitive conditions governing a bank's overall funding costs. Bank holding companies allocate resources across affiliated institutions through internal capital markets, allowing lending in one location to draw on funding raised elsewhere in the group (Campello, 2002). Interest expense sensitivity therefore reflects funding conditions across the bank's geographic network and wholesale markets, whereas competition for a particular borrower depends on locally available lending alternatives.[2] Figure 5 in the Appendix provides supporting evidence for our concentration measure: county-level branch-count HHI is positively and significantly associated with concentration measured using banks' shares of SBA loan dollars. SBA lending shares, however, reflect banks' pricing and credit-supply decisions together with borrower demand, making contemporaneous loan-share HHI potentially endogenous to the fixed–variable rate differential. Using one-year-lagged branch-count HHI mitigates this direct source of simultaneity by measuring established banking presence without incorporating current lending volumes. Its association with SBA lending concentration supports its relevance for the pricing analysis, while conditioning on $\beta_b^{\exp}$ accounts for differences in measured funding-cost sensitivity.

[2]The relevance of locality for lending is particularly strong in the SBA 7(a) market. Di and Pattison (2020) report that 71% of loans in their matched 2017 sample were extended to businesses located within ten miles of a branch of their lender.

2.2 Empirical Facts

We document two facts about the pricing and quantity of fixed-rate credit. First, banks with less rate-sensitive funding charge higher fixed-rate premiums. In the loan-level regressions, greater local banking concentration is associated with a larger fixed–variable rate spread, conditional on funding sensitivity and observed loan and bank characteristics. Second, fixed-rate loan stocks expand following monetary tightening, while variable-rate loan stocks initially contract. The expansion of fixed-rate lending is smaller at banks operating in more concentrated markets. Together, these patterns point to lending competition, alongside banks' funding characteristics, as a determinant of the pricing and provision of fixed-rate credit.

2.2.1 Fixed-rate loan pricing and market concentration

We measure the fixed-rate premium as the difference between the origination rates on fixed- and variable-rate loans. Panel (a) of Figure 1 shows that banks with lower interest expense sensitivity, β_b^{exp}, charge higher premiums: a one-standard-deviation increase in sensitivity is associated with a 5.3-basis-point lower premium. This relationship runs against the prediction from earnings hedging alone. When funding costs respond little to interest rates, fixed-rate lending can stabilise bank earnings. Encouraging borrowers to choose fixed-rate contracts requires a lower premium. Holding lending competition and borrower demand constant, this hedging incentive therefore predicts lower premiums at banks with less rate-sensitive funding. The negative relationship in the figure suggests that other pricing incentives vary with funding sensitivity.

Panel (b) points to lending-market concentration as a possible source of these incentives. Banks with less rate-sensitive funding also operate in more concentrated SBA lending markets: a one-standard-deviation increase in funding sensitivity is associated with a 0.043 decline in bank-level SBA loan-share HHI. This association suggests that stable funding may coincide with greater lending-market power. Banks with lending-market power can have an incentive to restrict fixed-rate lending and maintain a higher premium to increase expected earnings. To the extent that fixed-rate loans hedge their earnings exposure, restricting these loans sacrifices earnings stability. Lending-market power may therefore counteract the incentive to supply fixed-rate protection cheaply that arises from stable funding.

To investigate these associations, we relate loan-level origination rates to banks' funding sensitivity and lending-market concentration. Following Section 2.1.3, we use one-year-lagged branch-count HHI as a proxy for lending-market power conditional on funding sensitivity. Unlike the SBA loan-share measure in Figure 1(b), this measure does not incorporate contemporaneous lending volumes, mitigating the direct simultaneity between loan pricing and realised lending shares. The regressions describe conditional associations rather than causal

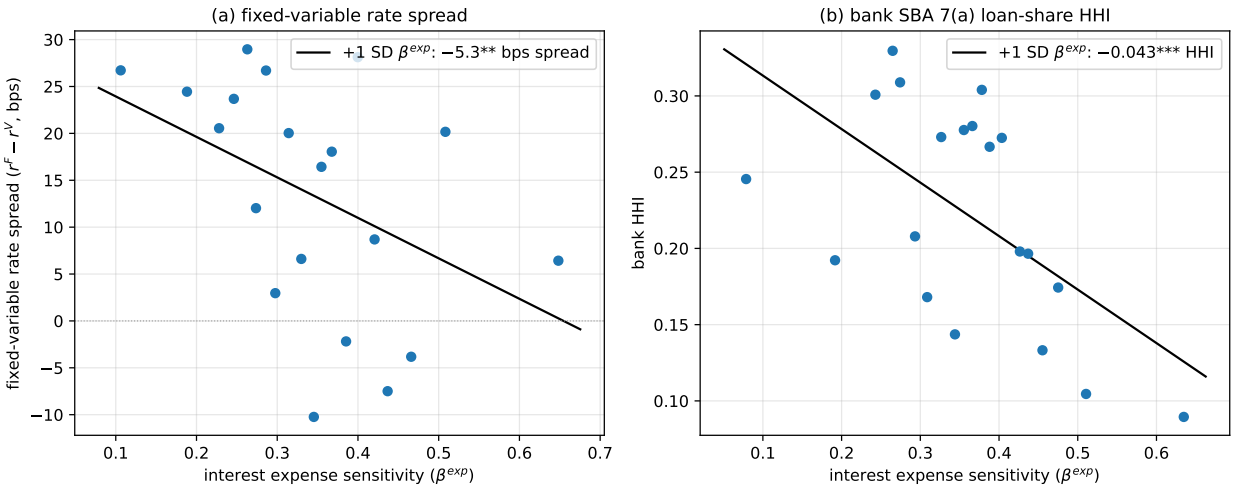


Figure 1: Bank funding sensitivity, fixed-rate premiums and lending concentration

Notes: Panel (a) plots the fixed–variable origination-rate differential against bank interest expense sensitivity. Panel (b) plots bank-level SBA 7(a) loan-share HHI against the same sensitivity measure. Dots represent binned observations, and solid lines show fitted relationships. Annotations report the change in each outcome associated with a one-standard-deviation increase in interest expense sensitivity. The loan-share HHI in Panel (b) differs from the lagged branch-count HHI used in the pricing regressions. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, and the Small Business Administration (SBA) 7(a) loan programme, 2010Q1–2019Q4.

effects. For loan i of rate type $k \in \{F, V\}$, issued by bank b in quarter t, we estimate

$$\begin{aligned} r_{i,k,b,t} = {} & \delta F_i + \alpha_\beta \beta_b^{\exp} + \alpha_H \mathrm{HHI}_{b,t-4} \\ & + d_\beta F_i \beta_b^{\exp} + d_H F_i \mathrm{HHI}_{b,t-4} \\ & + X_i'\gamma + F_i X_i'\gamma_F + W_{b,t}'\psi + F_i W_{b,t}'\psi_F \\ & + \eta_{\text{state},t} + \eta_{\text{industry}} + \varepsilon_i. \end{aligned}$$

Here, $F_i = \mathbf{1}\{k = F\}$ indicates a fixed-rate loan. Loan characteristics X_i include the guaranteed share, jobs supported and maturity, while bank characteristics $W_{b,t}$ include log assets and the equity-to-assets ratio. Their interactions with F_i allow these characteristics to have different associations with fixed- and variable-rate pricing. We include state-by-quarter and borrower-industry fixed effects and cluster standard errors by bank. The interaction coefficients d_β and d_H capture the relationships of funding sensitivity and concentration with the fixed–variable rate differential; the coefficients α_β and α_H capture their relationships with variable-rate pricing.

Table 1 reports estimates for 57,018 loans across 835 bank-year observations from 307 banks. The interaction between fixed-rate status and HHI is positive in all five specifications. In the preferred specification, Column (5), its coefficient is 3.866 with a standard error of 1.451, significant at the 1% level. A 0.10 increase in HHI is therefore associated with a 38.7-basis-point larger fixed–variable rate differential, holding funding sensitivity and the included

Table 1: Fixed- and variable-rate loan interest rates, interest expense sensitivity and one-year-lagged branch-count HHI

Dependent variable: interest rate at origination $r_{i,k,b,t}$ (%)	(1)	(2)	(3)	(4)	(5)
	OLS	quarter, industry FE	state×qtr, industry FE	loan-level controls	bank-level controls
Interest expense sensitivity (β_b^{exp})	-0.731	-0.252	-0.503*	-0.462	-0.193
	(0.506)	(0.292)	(0.285)	(0.291)	(0.303)
HHI, lagged one year ($\text{HHI}_{b,t-4}$)	-0.006	-0.044	-0.733	-0.976	0.154
	(1.394)	(0.803)	(0.861)	(0.910)	(0.952)
$F_i\times$ interest expense sensitivity	-0.197	-0.097	0.048	0.656	1.106
	(0.933)	(0.811)	(0.762)	(0.808)	(1.013)
$F_i\times$ HHI (lagged)	1.839	2.452	2.755	3.267*	3.866***
	(2.735)	(1.878)	(1.715)	(1.843)	(1.451)
Loan-level characteristics (X_i, F_iX_i)	No	No	No	Yes	Yes
Bank-level characteristics ($W_{b,t}$, $F_iW_{b,t}$)	No	No	No	No	Yes
Industry FE	No	Yes	Yes	Yes	Yes
State × quarter FE	No	No	Yes	Yes	Yes
Observations	57,018	57,018	57,018	57,018	57,018
Bank-year cells	835	835	835	835	835
Banks	307	307	307	307	307
R^2	0.011	0.382	0.432	0.463	0.481

Notes: standard errors clustered by banks in parentheses; *, **, ***: $p < 0.10, 0.05, 0.01$. Sample: SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks who had a branch network in the Summary of Deposits one year before the approval-year snapshot, restricted to bank × calendar-year cells with at least 2 fixed- and 2 variable-rate loans and fewer than 20% of their variable-rate loans at the statutory cap. The interest rate at origination, bank interest expense sensitivity, HHI and loan- and bank-level characteristics are winsorized at the 1st/99th percentiles within rate type. Interest expense sensitivity, β_b^{exp} and bank-level HHI, $\text{HHI}_{b,t}$ are constructed as described in Section 2.1.

characteristics constant. This finding is consistent with lending-market power supporting higher premiums for fixed-rate protection.

The preferred specification also yields a positive interaction between fixed-rate status and funding sensitivity: the coefficient is 1.106 with a standard error of 1.013. Its direction is consistent with the earnings-hedging mechanism. When funding costs are more rate-sensitive, an increase in those costs cannot be matched by repricing fixed-rate loan income, making fixed-rate lending less attractive to banks and increasing the premium they require. The estimate is not statistically distinguishable from zero, however, so the evidence for this conditional funding relationship remains suggestive.

For variable-rate loans, neither funding sensitivity nor concentration has a statistically detectable relationship with pricing in the preferred specification. The corresponding coefficients are -0.193 with a standard error of 0.303 for funding sensitivity and 0.154 with a standard error of 0.952 for HHI. Thus, the clearest pricing association is between concentration and the premium on fixed-rate loans relative to variable-rate loans.

Taken together, these facts highlight the relevance of relative market power in funding and lending for fixed-rate pricing. Stable funding creates an incentive to supply fixed-rate loans as an earnings hedge, while lending-market power creates an incentive to preserve premiums by restricting their supply. The concentration result and the direction of the conditional funding-sensitivity estimate are consistent with these opposing influences. They motivate a theoretical model in which banks weigh expected earnings against earnings stability, determining both the fixed-rate premium and the allocation of lending between fixed- and variable-rate contracts.

2.2.2 Fixed-rate and variable-rate lending

We next turn from pricing to quantities, examining how outstanding stocks of fixed- and variable-rate loans respond to monetary tightening and whether the fixed-rate response differs with lending-market concentration. We use the monetary policy shock series of Jarociński and Karadi (2020) and report responses to a contractionary shock scaled to 100 basis points.

Let $l^k_{b,t}$ denote the stock of loans of rate type $k \in \{F, V\}$ held by bank b at time t. We estimate cumulative responses separately by rate type using the local projection specification

$$\log l^k_{b,t+h} - \log l^k_{b,t-1} = \phi^k_h \epsilon^{MP}_t + W'_{b,t-1}\Gamma^k_h + Z'_{b,t-1}\Lambda^k_h + \alpha^k_b + \varepsilon^k_{b,t,h},$$

where h is the horizon in quarters and ϵ^{MP}_t is the monetary policy shock. The coefficient ϕ^k_h measures the cumulative response of loan stocks from the quarter preceding the shock to horizon h, conditional on the included controls. The vector $W_{b,t-1}$ contains bank characteristics such as asset size, leverage and funding sensitivity, while Z contains lagged aggregate

controls, including one-year Treasury yields, output growth and the yield-curve slope. Bank fixed effects are denoted by α_b^k. We report the responses by contract type and compare fixed-rate responses at banks with low and high HHI.

Panel (a) of Figure 2 shows opposite responses across contract types at short and medium horizons. Fixed-rate loan stocks rise by approximately 5.5% after four quarters and reach about 8% after ten quarters. Variable-rate loan stocks fall by roughly 5.5% after four to five quarters before recovering. The variable-rate point estimate turns positive around the ninth quarter, but the confidence bands widen substantially and include zero at longer horizons. The initial adjustment therefore shifts the composition of outstanding lending towards fixed-rate credit.

Panel (b) compares fixed-rate responses at banks with low and high bank-level HHI, our proxy for lending market power. The responses are similar on impact but diverge over subsequent quarters. At a ten-quarter horizon, the response is approximately 11% at low-HHI banks and 4.5% at high-HHI banks. The gap persists through the end of the twelve-quarter horizon. These point estimates complement the pricing evidence: more concentrated markets are associated with both a larger fixed–variable rate differential and a weaker expansion of fixed-rate lending following monetary tightening.

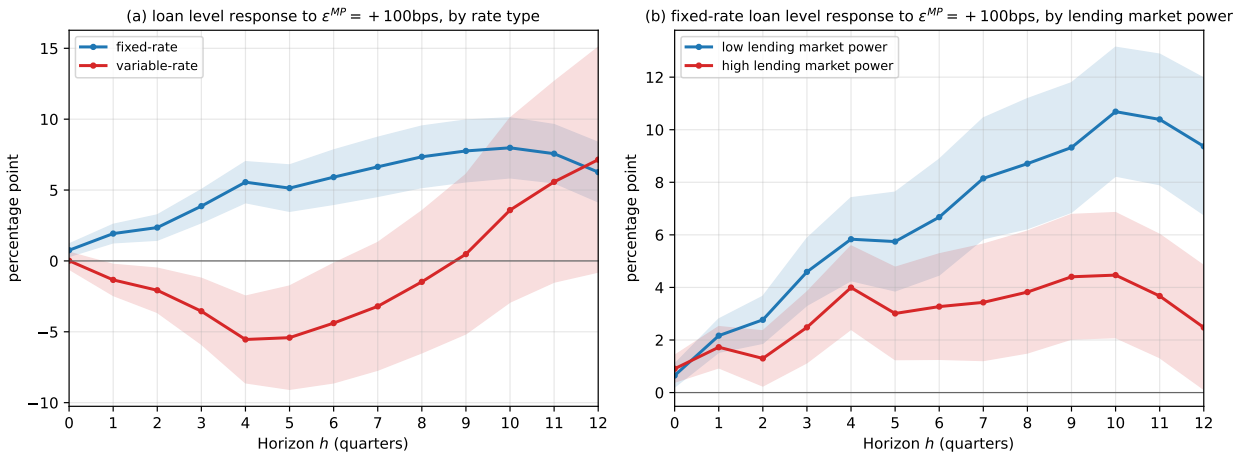


Figure 2: Fixed- and variable-rate lending following monetary tightening

Notes: Local-projection responses to a contractionary monetary policy shock from Jarociński and Karadi (2020), scaled to 100 basis points. Panel (a) compares fixed- and variable-rate loan stocks; Panel (b) compares fixed-rate loan stocks at banks with low and high lending market power, measured by bank-level HHI. Responses are expressed as 100 times cumulative log changes and can be read approximately as percentage changes in loan stocks. Horizons are in quarters, and shaded bands are 68% confidence intervals. Sources: FFIEC 031 Call Reports and FR Y-9C consolidated financial statements for bank holding companies, 2010Q1–2024Q4.

Loan stocks also reflect borrower demand and repayments, so their responses cannot be attributed to pricing alone. Nevertheless, monetary tightening shifts outstanding credit towards fixed-rate loans at short and medium horizons, with a weaker expansion at banks in more concentrated markets. These patterns complement the pricing evidence, suggest-

ing that relative market power in funding and lending shapes both the price of fixed-rate protection and the quantities of fixed- and variable-rate credit reaching the real economy.

These findings motivate a model in which funding and lending market power jointly determines the provision of fixed-rate loans. Stable funding creates an incentive to hold fixed-rate loans as an earnings hedge, encouraging banks to offer lower premiums. Lending market power creates an incentive to restrict fixed-rate credit and preserve high spreads. The higher relative fixed rates and weaker fixed-rate lending response observed at more concentrated banks are consistent with this pricing incentive limiting the supply of fixed-rate loans. The model in Section 3 formalises this tradeoff and examines how the relative strength of competition in funding and lending determines both borrowers' access to fixed-rate protection and the interest rate risk retained by banks. The link between pricing and loan quantities also motivates the quantitative extension of the core mechanism in Section 4, which assesses how bank risk management under market power in both funding and lending affects investment.

3 Model

In this two-date model, stable deposit costs give banks a reason to offer fixed-rate loans at a low premium: when funding costs move little with rates, fixed-rate income helps stabilise earnings. Lending power creates an opposing incentive. To attract more fixed-rate borrowers, a bank must lower the premium on its other fixed-rate loans. The portfolio that best hedges earnings can therefore differ from the one that maximises expected profits. We first hold bank counts fixed, then use entry to explain why deposit and lending competition move together.

3.1 Setup

Loans originate at $t = 0$ and pay at $t = 1$. The only aggregate uncertainty is the net benchmark rate,

$$r_1(z) = r_0 + \sigma z, \qquad \Pr(z = 1) = \Pr(z = -1) = \tfrac{1}{2}, \qquad \sigma > 0. \tag{1}$$

Write $x(z)$ for the outcome in state z. Expectations average over the two states: $\mathbb{E}[x(z)] = [x(-1) + x(1)]/2$. Thus $\mathbb{E}[r_1(z)] = r_0$ and $\mathrm{Var}(r_1(z)) = \sigma^2$. All contracts are chosen before z is realised. The parameters and distributions below are common knowledge.

Entrepreneurs. A unit mass of entrepreneurs each borrows one unit to finance a project with deterministic output, fixing total lending at one. Output net of principal repayment,

y, ensures participation and repayment in both states. Borrowers choose between fixed and variable rates. They receive no deposit income and have no other hedging instruments.

Entrepreneurs have mean–variance preferences over terminal wealth and heterogeneous risk aversion, $\tilde{\theta} \sim U[0, 2\theta]$, with $\theta > 0$. Each knows its type. A fixed-rate loan charges $r^F = r_0 + s$, where s is the premium set at origination; a variable-rate loan charges $r^V(z) = r_1(z)$. The pricing asymmetry reflects bank market power in fixed-rate loan market relative to variable-rate loan market.[3] Their certainty equivalents are

$$CE^F = y - r_0 - s, \qquad CE^V(\tilde{\theta}) = y - r_0 - \frac{\tilde{\theta}}{2}\sigma^2.$$

Borrowers choose a fixed rate when $CE^F > CE^V$, or equivalently, $\tilde{\theta} \geq \theta^* \equiv 2s/\sigma^2$. Rate-type sorting based on risk aversion resembles Campbell and Cocco (2003). For an interior allocation, fixed-rate demand and its inverse are

$$L^F = 1 - \frac{s}{\theta\sigma^2}, \qquad s(L^F, \sigma) = \theta\sigma^2(1 - L^F), \qquad \theta^* = 2\theta(1 - L^F). \tag{2}$$

Demand is one for $s \leq 0$ and zero for $s \geq \theta\sigma^2$; equilibrium will be interior. Expanding fixed-rate lending requires attracting less risk-averse borrowers, who will buy protection only at a lower premium. Throughout, $s = r^F - \mathbb{E}[r^V(z)]$ denotes the expected fixed-minus-variable premium.

Deposit and lending markets. There are $M \geq 1$ local deposit markets and a common lending market. With integer bank counts, each local market has n banks and the total is $N = Mn$. We allow $n \geq 1$ and $N = Mn$ to vary continuously as effective competition indices: n determines access to competing deposit offers, while $1/N$ measures an individual lender's influence on the fixed-rate premium. Integer values admit the symmetric bank-count interpretation; non-integer values use its continuous extension. Depositors compare offers locally, whereas borrowers access the common lending market. We initially treat n and M as primitives. Section 4 allows heterogeneous local markets and segments lending as well as deposits.

Total deposits are one, divided equally across banks in the integer benchmark, each with predetermined loan capacity $d_i = 1/N$. Depositors are risk-neutral and have an outside savings return $r_a > 0$. This exogenous reservation rate is known at origination and fixed across interest-rate states, competition indices, and market sizes. Its provider and pricing are outside the model. Households cannot directly access the wholesale benchmark $r_1(z)$. We assume $r_0 > r_a$: depositors compare expected returns, so the benchmark need not exceed

[3]The protection offered by fixed-rate financing has few accessible substitutes: Vickery (2008) documents financially constrained small firms disproportionately choose fixed-rate loans and seldom use derivatives to hedge interest rate risk. Public bond market is often costly to access for smaller firms (Faulkender, 2005).

r_a in every state. We focus on parameter configurations in which banks do not voluntarily increase deposit pass-through to hedge earnings. Appendix B.3 gives sufficient conditions and their compatibility with a negative premium–beta relationship.

Deposit search and pricing. Each depositor has an offer from its incumbent and compares it with a rival's offer if one arrives before the decision deadline. Measure search time in units that give the deadline an exponential arrival rate of one. Each of the $n-1$ rivals makes contact independently at rate $\zeta > 0$ in these units. Thus ζ measures the contact rate per rival relative to the deadline rate. With a continuum of accounts, the fraction receiving a competing offer is

$$p(n;\zeta) = \frac{\zeta(n-1)}{1+\zeta(n-1)}. \tag{3}$$

Throughout this section we assume equal contact and deadline rates, $\zeta = 1$, and write $p(n) = p(n;1) = 1 - 1/n$. Search is exogenous and independent of z and borrower types. The aggregate contact rate extends to real $n \geq 1$; the constant market-breadth threshold below uses $\zeta = 1$.

Eligible banks know whether a depositor has a competing offer and quote personalised indexed contracts $r_a + \alpha(r_1(z) - r_a)$, with $\alpha \geq 0$. Here α is account-level passthrough: $\alpha = 0$ pays r_a, while $\alpha = 1$ pays $r_1(z)$. Since $r_0 > r_a$, risk-neutral depositors prefer larger α and remain with the incumbent when expected returns tie, including a tie with outside savings. Quotation is costless and separate fees are excluded. Banks can replace a lost deposit by borrowing at $r_1(z)$, and invest a newly acquired surplus deposit at $r_1(z)$. This wholesale market accommodates deposit movements against committed loan books; households cannot access it directly.

Competition bids contested accounts up to $r_1(z)$, eliminating their funding advantage relative to wholesale borrowing. Switching at this rate would leave the incumbent's funding cost unchanged, while the receiving bank could invest the extra deposit at the same rate. In the maintained regime, uncontested accounts receive r_a. The tie-breaking rule keeps deposits with their incumbents: competition changes rates even without realised deposit flows. Appendix B.2 verifies optimal bids, allowing for their hedging benefits; $r_1(z)$ is an equilibrium bidding limit, not an imposed cap. Equilibrium average funding costs and passthrough are

$$r^d(z) = (1-\beta^d)r_a + \beta^d r_1(z), \qquad \beta^d \equiv \frac{\partial r^d(z)}{\partial r_1(z)} = p(n) = 1 - \frac{1}{n}.$$

More local competitors expose more accounts to rival offers, making funding costs more responsive to rates. As $n \to \infty$, banks pass through rate changes one-for-one, i.e. $\beta^d \to 1$. In contrast, when there is a monopoly bank, i.e. $n = 1$, the bank does not pass through any rate changes, i.e. $\beta^d = 0$. The exogenous r_a sets the funding-cost floor without directly

changing pass-through; a rate-sensitive outside return would change this conclusion. The construction uses personalised bids and observable comparison sets, rather than a Burdett–Judd posted-rate equilibrium.

Lending and bank preferences. In the integer benchmark, bank i chooses its fixed-rate portfolio share $l_i^F \in [0,1]$. Its fixed quantity and aggregate fixed lending are

$$q_i^F = d_i l_i^F = \frac{l_i^F}{N}, \qquad L^F = \sum_{i=1}^{N} q_i^F = \frac{1}{N}\sum_{i=1}^{N} l_i^F.$$

The remainder of its book is lent at the variable rate. Banks commit to fixed quantities and compete in Cournot fashion, internalising $\partial s/\partial l_i^F = -\theta\sigma^2/N$. The continuous extension retains this price-impact coefficient in the symmetric pricing conditions. All borrowers face the same fixed premium, while banks take $r^V(z) = r_1(z)$ as given. Borrower types can be allocated across banks to fill their capacities.

With retained deposits and bank-specific average pass-through β_i^d, net interest earnings per unit of assets are $\pi_i(z) = l_i^F r^F + (1 - l_i^F) r^V(z) - r_i^d(z)$. Substituting the contract rates gives

$$\pi_i(z) = (1 - \beta_i^d)(r_0 - r_a) + l_i^F s(L^F, \sigma) + (1 - \beta_i^d - l_i^F)\sigma z. \tag{4}$$

Principal repayments cancel. The sensitivity gap is

$$G_i \equiv \frac{\partial \pi_i(z)}{\partial r_1(z)} = (1 - l_i^F) - \beta_i^d,$$

so earnings variance is $\sigma^2 G_i^2$. The gap captures the difference between interest income and interest expense sensitivities. A positive gap means that higher rates raise bank earnings; a negative gap means that they lower earnings. Banks maximise

$$\max_{l_i^F \in [0,1]} \left\{ \mathbb{E}\pi_i(z) - \frac{\gamma}{2} \operatorname{Var}(\pi_i(z)) \right\}.$$

Throughout, γ applies to earnings per unit of assets (Hart & Jaffee, 1974). Credit losses, capital requirements, prepayment, separate origination fees, and additional hedging instruments are excluded.

Timing. Comparison sets are realised and observed by eligible banks before the rate shock. Banks then choose fixed-loan quantities, anticipating the deposit continuation. The fixed premium clears the loan market and entrepreneurs select contracts. Eligible banks subsequently quote indexed deposit contracts, depositors choose their accounts, and wholesale funding accommodates any movement. All contracts settle after z is realised. In the selected

continuation all deposits are retained and wholesale positions are zero.

3.2 Equilibrium

For given integer counts n, N, banks choose portfolios taking rivals' quantities as given and anticipating optimal deposit bids after every feasible portfolio profile. Borrowers and depositors choose their preferred available contracts, and loan and funding balances clear. We select the deposit retention continuation above and focus on symmetric lending, with $l_i^F = L^F$ and $G = 1 - \beta^d - L^F$. For non-integer competition indices, the corresponding symmetric conditions define the continuous extension.

Lemma 1 (Fixed-rate pricing in symmetric Cournot equilibrium)**.** *Fix n and N and suppose the deposit-retention continuation yields $\beta^d = 1 - 1/n$ at equilibrium. In the symmetric Cournot equilibrium, the fixed-rate premium and the common fixed-loan share $l_i^F = L^F$ satisfy*

$$s = \underbrace{\frac{\theta\sigma^2 L^F}{N}}_{\text{Cournot term}} - \underbrace{\gamma\sigma^2 G}_{\text{marginal hedging benefit}}. \tag{5}$$

Proof. See Appendix B.2. □

When funding costs respond less to rates than loan income ($G > 0$), replacing variable-rate loans with fixed-rate loans stabilises earnings. The bank will accept a lower premium for this hedge. Yet attracting another borrower requires a lower premium on all the fixed loans it originates. The Cournot term measures the premium revenue forgone on its other loans, which the bank weighs against the additional loan's income and hedging benefit. Banks choose fixed quantities at origination, when the common premium clears the market. Existing fixed loans retain their contracted rate.

With lending power, the bank can therefore prefer to supply less protection and tolerate earnings fluctuations in order to earn higher expected profits. Stable funding gives it a reason to supply protection cheaply, while lending power rewards restricting that supply. This hedging argument applies while $G > 0$: once fixed lending exceeds the full hedge, $G < 0$ and additional protection increases the bank's earnings risk, requiring compensation. The exogenous r_a affects expected funding costs but, conditional on bank counts, neither marginal incentive in (5).

Proposition 1 (Equilibrium and earnings exposure)**.** *Conditional on the deposit retention equilibrium, there is a unique symmetric lending equilibrium. Let $D = \theta(1 + 1/N) + \gamma$ and $\beta^d = 1 - 1/n$. Then*

$$L^F = \frac{\theta + \gamma(1 - \beta^d)}{D} = \frac{\theta + \gamma/n}{D}. \tag{6}$$

The corresponding premium is

$$s = \theta\sigma^2 \frac{\theta/N + \gamma\beta^d}{D}, \tag{7}$$

and earnings exposure is $G = \theta[1 - (N+1)\beta^d]/(ND)$. *The allocation is interior, with* $0 < L^F < 1$ *and* $0 < s < \theta\sigma^2$. *The sensitivity gap is positive, zero, or negative as* β^d *is below, equal to, or above* $1/(N+1)$.

Proof. Combine (2) and (5) to obtain $DL^F = \theta + \gamma(1 - \beta^d) > 0$. The difference between D and this numerator is $\theta/N + \gamma\beta^d > 0$, establishing interiority. Strict own concavity makes the first-order condition sufficient. The symmetric linear equation has one solution, while the symmetric all-zero and all-one portfolios fail the boundary optimality conditions. Substitution gives the premium and the gap. □

Two portfolio targets explain the allocation. Full earnings hedging requires $L_H = 1 - \beta^d = 1/n$, while a risk-neutral bank chooses $L_C = N/(N+1)$. Equilibrium satisfies

$$L^F = \omega L_H + (1 - \omega) L_C, \qquad \omega = \frac{\gamma}{D} \in (0, 1).$$

When $L_H > L_C$, preserving lending rents leads banks to stop short of the full hedge: they retain $G > 0$ and forgo some profit stability. When $L_H < L_C$, earning premium income instead draws them beyond the full hedge, leaving $G < 0$. In both cases, banks accept earnings risk to raise expected profits relative to the perfectly hedged portfolio. Greater risk aversion raises the weight on L_H, but complete hedging occurs only when the two targets coincide.

Greater rate uncertainty σ raises the premium but leaves the fixed share unchanged. Under mean–variance preferences, borrowers' willingness to pay and banks' compensation for bearing risk both scale with variance. Other preferences or endogenous participation (Section 3.6) can change the share response.

3.3 Competition and the premium–beta relationship

Deposit and lending competition change different sides of this tradeoff. More responsive funding reduces the fixed-rate share needed to stabilise earnings, weakening banks' willingness to supply borrower protection. More lenders instead reduce each bank's incentive to restrict supply to preserve the premium.

First vary deposit pass-through and lending competition separately. Although β^d depends on n in equilibrium, treating it as exogenous lets us isolate the funding effect. Holding the other margin fixed gives

$$s_{\beta^d} = \frac{\theta\gamma\sigma^2}{D} > 0, \qquad s_N = -\frac{\theta^2\sigma^2[\theta + \gamma(1 - \beta^d)]}{N^2 D^2} < 0.$$

Since $d\beta^d/dn = 1/n^2 > 0$, more local deposit competition alone raises the premium. Therefore, *a negative premium–beta relationship requires lending competition to increase enough to offset this funding effect.*

To study joint changes in deposit and lending competition, hold fixed the ratio of lenders to local deposit competitors,

$$M \equiv \frac{N}{n} \geq 1, \qquad N = Mn = \frac{M}{1-\beta^d}.$$

Here M is the number of local deposit markets accessible to borrowers, held fixed in this comparison.

Proposition 2 (Market breadth and the premium slope)**.** *Along symmetric configurations with a fixed ratio $M = N/n$, the fixed premium is decreasing in deposit beta if*

$$M = \frac{N}{n} < \frac{\theta^2}{\gamma(\theta+\gamma)},$$

increasing if the inequality is reversed, and constant at equality. Under the uniform sufficient condition in Appendix B.3, the sign holds over $\beta^d \in [0,1)$ and orders the corresponding integer-count configurations.

Proof. Writing $g = \gamma/\theta$, substitution into (7) gives

$$\frac{s(\beta^d; M)}{\theta\sigma^2} = \frac{1 + (gM-1)\beta^d}{1 + M(1+g) - \beta^d}. \tag{8}$$

Differentiating (8) at fixed M yields

$$\frac{ds}{d\beta^d} = \theta\sigma^2 \frac{M[gM(1+g)-1]}{[1+M(1+g)-\beta^d]^2}.$$

The denominator is positive, giving the result. □

When borrowers rely on a narrow lending market, entry can substantially reduce the rents banks preserve by restricting fixed lending. The resulting expansion of protection can outweigh the loss of the hedging incentive as funding becomes more rate-sensitive. When borrowers already access many lenders, there is less lending power left to erode, so the funding effect dominates. A negative slope is feasible for $M \geq 1$ only if $\gamma/\theta < (\sqrt{5}-1)/2$. Figure 3 illustrates both regimes; the premium itself remains positive.

Price-taking lending. Let $N \to \infty$ while holding n, and hence deposit passthrough β_d, fixed. This case has no lending market power. The Cournot term in (5) disappears and (7)

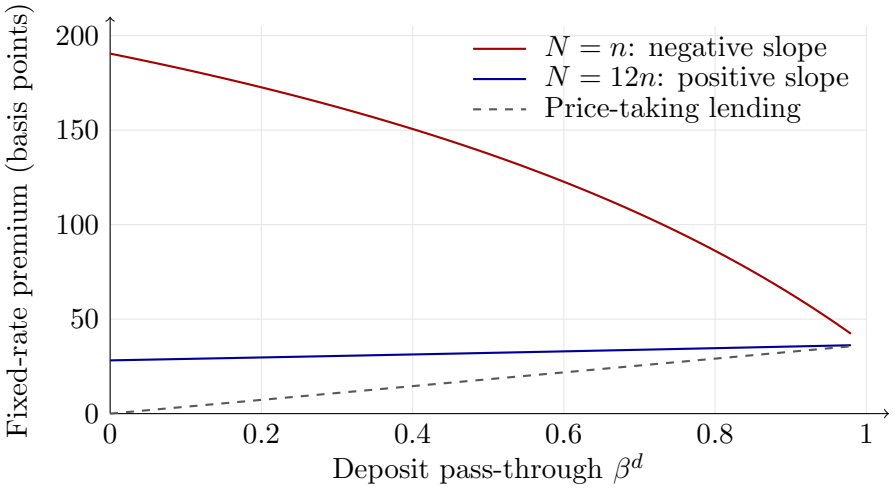


Figure 3: Fixed-rate premiums and deposit pass-through. The illustration uses $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, and $r_0 - r_a = 0.03$, giving a breadth threshold of 9.09 and satisfying the deposit-pricing condition in Appendix B.3. Solid curves hold N/n fixed and vary the continuous competition indices; the dashed curve takes $N \to \infty$ at a fixed beta. The curves are theoretical illustrations, not estimates.

implies

$$s_\infty(\beta^d) = \frac{\theta\gamma\sigma^2}{\theta + \gamma}\beta^d.$$

Without the incentive to restrict supply, stable funding translates into cheaper protection: the premium falls to zero as $\beta^d \to 0$. For positive beta, a positive premium compensates banks for taking fixed lending beyond their full hedge. This limit holds deposit competition fixed; taking $n \to \infty$ at fixed M also sends beta to one. Deposit rents can survive price-taking lending because banks choose the composition of predetermined books.

3.4 The premium–beta relationship under CARA and CRRA

The fixed-rate premium can fall as deposit pass-through rises under both constant absolute and constant relative risk aversion. As in Section 3.3, more responsive deposit costs weaken a bank's incentive to hedge with fixed-rate loans. But when borrowers gain access to more lenders, each bank also has less power to sustain a high premium. To establish when lending competition dominates, keep the same deposit contracts and borrower types and let borrowers and banks maximise expected utility.

With a slight abuse of notation, retain $\tilde{\theta} \sim U[0, 2\theta]$ for borrower risk aversion and γ for bank risk aversion. For a borrower set $\varrho = \tilde{\theta}$ and for a bank set $\varrho = \gamma$. Consider two utility functions over terminal wealth w:

$$u_\varrho^{\text{CARA}}(w) = \frac{1 - e^{-\varrho w}}{\varrho}, \qquad u_\varrho^{\text{CRRA}}(w) = \begin{cases} (w^{1-\varrho} - 1)/(1 - \varrho), & \varrho \neq 1, \\ \log w, & \varrho = 1. \end{cases}$$

Under constant absolute risk aversion (CARA), ϱ measures aversion to a given dollar risk. Under constant relative risk aversion (CRRA), it measures aversion to a given proportional risk, and w must be positive. The parameters therefore have different units across specifications. In either case $\varrho = 0$ gives linear utility, taking the limit for CARA.

As before, the most risk-averse borrowers choose fixed rates. At a fixed share L, the marginal borrower has type $2\theta(1-L)$. Let $c = y - r_0 > \sigma$ be its consumption after paying the expected variable rate. Inverse demand $s(L, \sigma)$ is the premium at which this borrower is indifferent between consuming $c - s$ for certain and consuming $c - \sigma$ or $c + \sigma$ with equal probability.

For both preferences, let the bank start with the same predetermined equity E per unit of assets and evaluate terminal equity $E + \pi_i(z)$. At a symmetric allocation its earnings exposure is $G = 1 - \beta^d - L$, and its terminal equity per unit of assets is $W \pm \sigma G$, where $W = E + (1 - \beta^d)(r_0 - r_a) + Ls$. Under CARA, adding the same equity in both states leaves their relative marginal utilities unchanged. Under CRRA, a given earnings fluctuation matters more when equity is low. We choose E so that terminal equity is positive in both states for every feasible portfolio; $E = 0$ also suffices if earnings alone meet this condition.

Lemma 2 (Fixed-rate pricing under expected utility)**.** *In an interior symmetric Cournot equilibrium, the fixed-rate premium satisfies*

$$s = \underbrace{-\frac{L}{N} s_L}_{\text{Cournot term}} - \underbrace{H}_{\text{marginal hedging benefit}}, \qquad H = \sigma \frac{u'_\gamma(W - \sigma G) - u'_\gamma(W + \sigma G)}{u'_\gamma(W - \sigma G) + u'_\gamma(W + \sigma G)}, \tag{9}$$

where $s_L = \partial s(L, \sigma)/\partial L$ and u_γ is the bank's utility under the specification considered.

Proof. See Appendix B.4. □

Consider a risk-neutral bank taking rivals' quantities as given. An additional fixed loan earns the premium s but lowers the premium on the bank's other fixed loans. Since $\partial L/\partial l_i^F = 1/N$, its first-order condition is $s + (l_i^F/N)s_L = 0$. At the common equilibrium share L_C, this becomes

$$s(L_C, \sigma) + \frac{L_C}{N} s_L(L_C, \sigma) = 0. \tag{10}$$

Now consider a risk-averse bank at that same portfolio. If $L_C < 1 - \beta^d$, its earnings are lower in the low-rate state. An additional fixed loan shifts income into that state. Its marginal expected profit is zero at L_C, but its hedging benefit is positive, so the bank wants to expand fixed lending. If $L_C > 1 - \beta^d$, the same loan adds to earnings risk and the bank wants to supply less. Under CARA the earnings gap determines the value of this hedge. Under CRRA a bank with less equity values the same reduction in risk more.

Proposition 3 (Portfolio choice under expected utility)**.** *Let L_C denote the symmetric Cournot fixed share with risk-neutral banks that solves* (10)*, and $L_H = 1 - \beta^d$ the share that fully hedges bank earnings. Under CARA the symmetric Cournot equilibrium is unique. Its fixed share lies between L_C and L_H, strictly so when they differ. The same result holds under CRRA if premium revenue is strictly concave in the bank's own fixed share, symmetric marginal revenue is strictly decreasing in L, and G/W decreases in L.*

Proof. See Appendix B.4. □

To compare markets, again hold $M = N/n$ fixed. As deposit beta rises, the bank needs fewer fixed loans to hedge deposits. At the same time, borrowers gain access to more lenders, reducing the premium revenue each bank gives up when it expands fixed lending.

Proposition 4 (Market breadth and the premium slope under expected utility)**.** *Along $N = Mn$, suppose the marginal return $s + (L/N)s_L + H$ is strictly downward sloping in L at the symmetric equilibrium. The fixed-rate premium decreases with deposit beta if and only if*

$$\frac{L(-s_L)}{M} > -\left.\frac{\partial H}{\partial \beta^d}\right|_L.$$

Under the conditions of Proposition 3, if symmetric marginal revenue has strictly negative slope at L_C, this inequality holds for sufficiently small positive γ under either utility function.

Proof. See Appendix B.4. □

The left side measures how much entry weakens the incentive to restrict fixed lending. The right side measures how much more responsive deposits weaken the incentive to hedge. When banks place little weight on earnings risk, the pricing incentive dominates even though banks still value a hedge.

In the CRRA illustration, doubling n from two to four raises deposit beta from 0.50 to 0.75. Only market breadth differs across economies. Set $\theta = 2$ (borrower types range from zero to four), $\gamma = 0.80$, $c = 0.08$, $E = 0.10$, and $r_0 - r_a = 0.03$. A larger shock, $\sigma = 0.02$, makes the premium changes easier to see.

With two deposit markets ($M = 2$), borrowers gain access to eight lenders instead of four ($N = Mn$). The premium falls from 25.66 to 24.50 basis points and the fixed share rises from 74.75% to 75.89%. With twelve deposit markets ($M = 12$), borrowers already face 24 lenders before the increase and 48 afterwards. The premium rises from 13.73 to 18.45 basis points and the fixed share falls from 86.49% to 81.86%.

Where borrowers initially have few lenders, competition makes protection cheaper despite weaker hedging incentives. In a broad market, lenders have less pricing power to lose and the funding effect can raise the premium.

3.5 Welfare and the allocation of interest-rate risk

Deposit competition determines how closely funding costs follow market rates. A bank with stable deposit costs can insure borrowers against a rate increase at little risk to its earnings. Lending competition determines how much of that protection reaches borrowers: selling another fixed loan requires a bank with pricing power to lower the premium on its other fixed loans.

Compare efficient and Cournot lending for the same deposit contracts, bank counts, loan menu, and total credit. Premium payments transfer income from borrowers to banks. The loss arises when a borrower retains rate risk even though fixing the loan would raise joint surplus.

Let L be the share of loans allocated to fixed-rate borrowers, starting with the most risk averse. Borrowers with $\tilde{\theta} < 2\theta(1-L)$ retain variable-rate loans, and each bank retains an earnings exposure of $1-\beta^d-L$. Summing borrower and depositor certainty equivalents and weighting banks' per-asset certainty equivalents by their asset shares gives, up to a constant,

$$\mathcal{W}(L) = -\frac{\sigma^2}{2}\left[\theta(1-L)^2 + \gamma(1-\beta^d-L)^2\right]. \tag{11}$$

The first term measures the cost of rate risk borne by borrowers and the second the cost borne by banks. Risk-neutral depositors require no compensation for risk. The efficient share,

$$L^\star = \frac{\theta + \gamma(1-\beta^d)}{\theta+\gamma},$$

gives more weight to insuring borrowers when θ is high and to hedging bank earnings when γ is high. At the Cournot allocation, $\mathcal{W}'(L^F) = \theta\sigma^2 L^F/N > 0$: the gain from one more fixed loan equals the premium revenue the bank would forgo on its other fixed loans. By (11), the loss from this shortfall is

$$\mathcal{W}(L^\star) - \mathcal{W}(L^F) = \frac{\sigma^2}{2}(\theta+\gamma)(L^\star - L^F)^2.$$

Under mean–variance preferences, greater uncertainty raises both parties' risk costs in proportion to σ^2. The efficient and equilibrium shares therefore stay fixed, and doubling σ quadruples the welfare loss for any γ/θ.

Uncertainty and risk aversion. With CARA preferences, a more risk-averse borrower will pay more to fix the rate. What makes it profitable for a bank to withhold a loan, however, is the difference between what it can charge its current borrowers and what it must charge to attract the next one. For small shocks, marginal willingness to pay is approximately $\theta\sigma^2(1-L)$. A higher θ then steepens inverse demand: expanding fixed lending requires

a larger premium cut on the bank's existing fixed volume. This is one way borrower risk aversion can make lending power more costly.

Proposition 5 (Uncertainty and the welfare cost of lending power)**.** *Under CARA preferences with $\theta, \gamma > 0$, fix deposit pass-through $0 < \beta^d < 1$ and a finite number of lenders. Let $\Delta(\sigma)$ be the gain in aggregate monetary certainty equivalents from replacing symmetric Cournot lending with efficient lending. The gain is positive for every $\sigma > 0$ and attains its maximum at an interior level of uncertainty. It tends to zero as $\sigma \to 0$ and as $\sigma \to \infty$.*

Proof. See Appendix B.5. □

The loss therefore has an interior peak. At low uncertainty, an increase in σ makes each loan left unhedged more costly. At high uncertainty, even the marginal fixed-rate borrower will pay close to σ for protection. Borrowers' valuations converge relative to the size of the shock, while banks place increasing weight on losses in their adverse state. Both allocations approach the bank's full hedge, $1-\beta^d$, and the additional loss due to lending power vanishes. This large-shock comparison holds deposit contracts fixed.

For $n = N = 2$ and $\beta^d = 1/2$, raise σ from 0.005 to 0.015. At $\theta = 1600$, the loss rises from 2.74 to 4.14 basis points per unit lent for $\gamma = 80$ but falls from 1.24 to 0.46 for $\gamma = 320$. Borrower preferences can also reverse the comparison. At $\gamma = 160$, the loss rises from 2.08 to 2.12 basis points for $\theta = 800$ but falls from 2.11 to 1.91 for $\theta = 1600$, even though these borrowers value protection more.

3.6 Market size and endogenous entry

Section 3.3 shows when the premium falls as bank counts rise. What makes depositors and borrowers face more competitors in the first place? A larger customer base can cover the fixed cost of another bank. Because each entrant takes deposits locally and lends across communities, one participation decision expands choice on both sides of the market.

Let $A > 0$ denote both the mass of unit loans and total deposits. Each of the M communities supplies A/M deposits. Search remains uncongested, so market size affects pass-through only through entry. For this extension assume $r_0 - r_a > \gamma\sigma^2$, which rules out voluntary extra pass-through after any portfolio choice and supports the franchise-value inequalities below. Earlier results require only the weaker condition in Appendix B.3.

To operate, owners pay $F > 0$ in terminal goods from outside resources. This cost is independent of market size and beta. Staying out pays zero. Before search, active banks share home-community deposits equally, extending the benchmark allocation without a separate franchise acquisition price. We study configurations with at least one bank per community.

Under symmetry, $N = Mn$ banks each hold $A/(Mn)$ in assets. Fixed volume is AL^F, so inverse demand (2) and all per-asset outcomes carry over. Risk-adjusted value per unit of

assets before entry costs is

$$W(n;M) = \frac{r_0 - r_a}{n} + L^F(n;M)s(n;M) - \frac{\gamma}{2}\sigma^2\left[\frac{1}{n} - L^F(n;M)\right]^2, \tag{12}$$

and $v(n;M) = W(n;M)/(Mn)$ gives bank value per unit of market size. Entry yields $Av(n;M) - F$. At given counts a higher outside return leaves portfolios unchanged but erodes deposit rents, $\partial W/\partial r_a = -1/n$. We hold r_a fixed.

We use a continuous bank count to trace how participation responds to market size. Appendix B.7 gives the exact entry conditions for integer counts in one or several communities.

Proposition 6 (Market size and participation)**.** *Under the maintained parameter restrictions, $W(n;M) > 0$ and $W_n(n;M) < 0$ for every real $n \geq 1$ and $M \geq 1$. Consequently $v(n;M)$ is strictly decreasing and tends to zero.*

For any fixed $M \geq 1$, the continuous symmetric participation equation

$$\frac{A}{Mn}W(n;M) = F \tag{13}$$

has a unique solution $n > 1$ whenever $AW(1;M)/M > F$. Along this approximation,

$$\frac{dn}{dA} = \frac{n/A}{1 - nW_n/W} > 0, \qquad \frac{d\beta^d}{dA} = \frac{1}{n^2}\frac{dn}{dA} > 0, \qquad \frac{dN}{dA} = M\frac{dn}{dA} > 0. \tag{14}$$

The premium–beta slope is again given by Proposition 2.

Proof. See Appendix B.6. □

Entry divides the larger market among more banks. More depositors receive competing offers and more lenders compete for fixed-rate borrowers. Both margins erode rents, limiting further entry.

At fixed M, market size thus selects a point on the premium–beta curve in Section 3.3. Larger markets have higher beta and cheaper protection when competition for borrowers dominates. When more responsive funding matters more for hedging, beta and the premium rise together. Proposition 2 gives the threshold.

4 Loan pricing, investment, and the macroeconomy

In Section 3, each entrepreneur borrows one unit. Here a borrower facing a higher fixed premium can switch to floating finance or invest less. Both choices reduce fixed lending, but only the change in project scale directly changes demand for labour. Household saving finances these projects, linking investment to current consumption. We study this allocation

in general equilibrium conditional on the benchmark rate process and the outside savings return.

Environment. There are M markets. Market m has a mass φ_m of entrepreneurs, with $\sum_{m=1}^{M} \varphi_m = 1$. As in Section 3.6, we use a continuous count n_m for active banks and write $N = \sum_m n_m$. Here each bank takes deposits and lends in its own market. Borrowers can approach any local bank. Depositors search as in (3), with local contact rate ζ_m relative to the decision deadline. Under the same deposit-pricing rule,

$$r_m^d(z) = (1 - \beta_m^d) r_a + \beta_m^d r_1(z), \qquad \beta_m^d = p(n_m; \zeta_m). \tag{15}$$

The real interbank rate follows (1). We vary r_0 holding σ, r_a, search frictions and participation cost schedules fixed. A government facility can implement the benchmark by intermediating offsetting bank positions at the same floating rate, with zero net borrowing and taxes (Appendix C.2).

Retailers contract their date-0 supply prices before the change in r_0 and price flexibly at date 1. Banks observe r_0 and decide whether to serve a market before deposit search and lending. Loan terms, investment, employment at both dates and the terminal wage are chosen before the rate fixing. Contracts settle at date 1.

4.1 Households

A representative household saves through established deposit relationships. Of total saving $\mathcal{D}$, market m receives $\mathcal{D}_m = \varphi_m \mathcal{D}$ and each local bank receives $d_m = \mathcal{D}_m / n_m$. The household chooses how much to save, taking these shares as given. Its utility is

$$\mathcal{U} = \log C_0 - \frac{\chi_0 L_0^{1+\nu_0}}{1+\nu_0} + \delta_H \left(\mathbb{E}[C_1(z)] - \frac{\chi_1 L_1^{1+\nu_1}}{1+\nu_1} \right). \tag{16}$$

Here $\delta_H \in (0,1)$ discounts terminal utility and $\chi_t, \nu_t > 0$ govern labour costs. Linear terminal utility is a useful approximation when fluctuations in deposit interest are small relative to the family's consumption. It lets depositors rank offers by expected repayment as in Section 3.1, while log utility at date 0 preserves a saving decision sensitive to current consumption.

The household owns retailers and receives their profits Ω_t^R at both dates, taking dividends as given in its individual choices. Competitive intermediate producers earn zero profits at date 0. With $R^d(z) = \sum_m \varphi_m [1 + r_m^d(z)]$, the date-0 budget is

$$C_0 + \mathcal{D} = w_0 L_0 + \Omega_0^R. \tag{17}$$

At maturity the household receives the return on those deposits as well as wages and retail

dividends:

$$C_1(z) = w_1 L_1 + R^d(z)\mathcal{D} + \Omega_1^R, \qquad z \in \{-1, 1\}. \tag{18}$$

Entrepreneurs consume their project profits and bank owners consume their banks' profits. Keeping these claims with their respective owners makes each bear the earnings risk that enters its financing or lending decision. Only retail dividends enter the saving household's budget. The household's saving decision satisfies

$$\frac{1}{C_0} = \delta_H \, \mathbb{E}[R^d(z)]. \tag{19}$$

At each date the household equates the marginal cost of labour to the value of its wage:

$$\chi_0 L_0^{\nu_0} = w_0/C_0, \qquad \chi_1 L_1^{\nu_1} = w_1.$$

A higher expected deposit return lowers current consumption at an interior solution. The contracted wage determines terminal labour supply.

4.2 Production, retail pricing and investment

One unit of date-0 labour produces A_0 intermediate goods at price $p_0 = w_0/A_0$. Monopolistically competitive retailers convert these goods one for one into retail varieties. With elasticity of substitution $\varepsilon > 1$, flexible terminal prices imply $p_1 = \bar{p} = (\varepsilon - 1)/\varepsilon$. Preset current retail prices let p_0 and current employment adjust after a rate change. Retail dividends at either date are

$$\Omega_t^R = (1 - p_t) Y_t. \tag{20}$$

Appendix C.2 derives retail pricing, including the baseline condition $p_0 = \bar{p}$.

An entrepreneur borrows the full cost of investment k, hires e units of labour and produces $Ak^\alpha e^{1-\alpha}$ intermediate goods. Capital depreciates fully and expansion costs $\kappa k^2/2$ terminal goods. At wage w_1, labour demand is $e = \ell(w_1)k$, where $\ell(w_1) = [(1-\alpha)\bar{p}A/w_1]^{1/\alpha}$. The return per unit invested after wages and principal repayment is

$$a(w_1) = \alpha \bar{p} A \ell(w_1)^{1-\alpha} - 1. \tag{21}$$

Write $\mu = a(w_1) - r_0$. Borrowers have CARA utility with $\log \tilde{\theta} \sim N(\log \theta - \tau^2/2, \tau^2)$, common across markets. On the interior branch, fixing at $r_0 + s_m$ gives investment $k_F = (\mu - s_m)/\kappa$. With floating finance, each extra unit borrowed increases exposure to the rate shock. Optimal investment solves

$$\kappa k_V + \sigma \tanh(\tilde{\theta} \sigma k_V) = \mu.$$

Borrowers compare the two contracts at their respective optimal scales. The cutoff θ_m^* equates their certainty equivalents, and types above it fix. A higher premium reduces the size of fixed-rate projects and leads some borrowers to switch contracts. Appendix C.1 derives the scale and cutoff conditions.

4.3 Bank participation and the price of protection

Each local bank chooses fixed credit taking rivals' quantities, deposits and wages as given. Floating loans and interbank positions earn $r_1(z)$, and $q_{im}^F + q_{im}^V + x_{im} = d_m$, where $x_{im} > 0$ denotes net interbank lending. Banks can borrow or lend freely at the benchmark and have no outside fixed-rate hedge. All have CARA utility over earnings per unit of deposits with constant coefficient γ. Let $s_m(Q;\mu) = s(Q/\varphi_m;\mu)$ denote inverse demand for local fixed credit. In symmetric Cournot equilibrium the premium satisfies

$$s_m = -\frac{Q_m^F}{n_m} s_{m,Q}(Q_m^F;\mu) - \sigma \tanh\left[\gamma\sigma\left(1 - \beta_m^d - \frac{Q_m^F}{\mathcal{D}_m}\right)\right].$$

Expanding fixed lending stabilises earnings when floating income is more rate-sensitive than deposit costs. A bank weighs that gain against the lower premium on its other fixed loans. The resulting premium enters both borrowers' contract choice and their investment decision. Appendix C.1 gives the bank's objective and optimality condition.

To decide whether to enter, a bank anticipates these earnings on its local deposit base d_m. Write $W_m = -\gamma^{-1}\log\mathbb{E}[\exp(-\gamma\pi_{im}(z))]$ for the certainty equivalent of operating earnings per unit of deposits, where $\pi_{im}(z) = (1-\beta_m^d)[r_1(z) - r_a] + (q_{im}^F/d_m)[r_0 + s_m - r_1(z)]$. Rank potential participants by the fixed fee they must pay to serve the market. On the interior branch the marginal bank breaks even:

$$d_m W_m = F_m n_m^{\kappa_E}. \tag{22}$$

The fee level $F_m > 0$ can differ across markets. The elasticity $\kappa_E > 0$ is common: it measures how much more costly it is to attract another local lender as participation expands. Fees are paid at settlement to owners in the financial sector and enter their aggregate consumption as transfers.

A higher r_0 widens the expected margin on existing deposits by $(1-\beta_m^d)$ per unit increase in the benchmark. This encourages entry where deposit rates adjust slowly. Entry in turn divides local deposits among more banks and raises deposit pass-through. On the lending side, each bank serves a smaller share of borrowers and sacrifices less premium income on its existing loans when it expands fixed lending. The equilibrium premium reflects both this change in competition and the change in the bank's incentive to hedge.

4.4 Aggregation and equilibrium

Let $\theta_m^* = \theta^*(s_m)$ and $f_m^F = 1 - F_\theta(\theta_m^*)$ be local fixed-contract take-up. Fixed lending equals borrower mass times fixed take-up times loan size:

$$Q_m^F = \varphi_m f_m^F \frac{\mu - s_m}{\kappa}.$$

Floating loan sizes vary across borrower types, so $Q_m^V = \varphi_m \int_0^{\theta_m^*} k_V(t)\,\mathrm{d}F_\theta(t)$. Together these loans finance local investment $I_m = Q_m^F + Q_m^V$. The interbank market is closed and there is no outside asset: $B^V = 0$. Under local symmetry, net interbank lending is $X_m = n_m x_{im}$. Write $J_2 = \sum_m \varphi_m \int_0^\infty k_m(t)^2\,\mathrm{d}F_\theta(t)$ for the aggregate second moment of investment. Let $C_1^E(z)$ and $C_1^B(z)$ denote aggregate terminal consumption of entrepreneurs and financial owners, respectively, and $C_1(z)$ household consumption. Market clearing requires

$$I = \sum_{m=1}^{M} I_m = \mathcal{D}. \tag{23}$$

Each market lends its excess deposits, $X_m = \mathcal{D}_m - I_m$, and interbank positions sum to zero. Current output finances consumption and investment:

$$Y_0 = C_0 + I = A_0 L_0. \tag{24}$$

Project labour demand determines terminal employment,

$$L_1 = \ell(w_1) I. \tag{25}$$

Terminal output covers consumption and project expansion costs in each state:

$$Y_1 = C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2, \qquad z \in \{-1, 1\}. \tag{26}$$

Since $Y_1 = A\ell(w_1)^{1-\alpha} I$, investment determines terminal output at a given wage. Repayments divide that output among households, entrepreneurs and bank owners; expansion costs absorb real resources. Appendix C.3 verifies the statewise accounts. The fixed share of new credit, $L^{F,\mathrm{agg}} = \sum_m Q_m^F / I$, weights projects by loan size, whereas $f^{F,\mathrm{agg}} = \sum_m \varphi_m f_m^F$ counts borrowers.

Definition 1 (General equilibrium conditional on the benchmark and outside rates)**.** *Given the rate process, outside return, market sizes, search frictions, participation cost schedules and inherited retail contracts, an interior equilibrium consists of local bank counts, prices, wages, consumption, employment, investment and financial portfolios such that:*

1. *The household maximises* (16) *subject to* (17) *and* (18)*, taking prices, dividends and*

deposit shares as given.

2. *Entrepreneurs choose contract, scale and committed employment. Producers, retailers and the final-good aggregator optimise at inherited current retail prices and flexible terminal prices.*

3. *In each market $n_m > 1$ satisfies* (22) *at the anticipated lending equilibrium. Given n_m, deposit rates follow* (15) *and banks play symmetric Cournot best responses taking deposits, wages and rival quantities as given, with $q_{im}^F = Q_m^F/n_m$ and $q_{im}^V = Q_m^V/n_m$.*

4. *Bank balance sheets and loan markets clear, interbank positions sum to zero, and* (23), (24), (25) *and* (26) *clear funding, goods and labour markets. Consumption and repayments are feasible in both states.*

Under local symmetry each lender originates a share $1/n_m$ of local credit, so the lending Herfindahl index is $1/n_m$.

5 Quantitative results

We calibrate the joint lending, household saving and investment equilibrium of Section 4. Households choose aggregate deposits, entrepreneurs choose both loan contracts and project sizes, and banks enter local markets. The policy-rate experiment and the risk-bearing-capacity application use the same parameters and preferences.

5.1 Specification and calibration

The three markets are ordered by their baseline deposit betas. Entrepreneur population shares satisfy $\varphi_1 \geq \varphi_2 \geq \varphi_3$ and sum to one. Deposit relationships have equal shares, $\mathcal{D}_m = \mathcal{D}/3$, but total household deposits $\mathcal{D}$ are endogenous. Baseline investment and deposits are normalised to one; after a shock, market clearing requires $\mathcal{D} = \sum_m I_m$, not $\mathcal{D} = 1$.

Financial preferences and entry. Borrowers have exact CARA utility over profit, augmented by a common fixed-contract convenience benefit $v_F k$. Market-specific lognormal risk-aversion distributions have medians $\exp(\mu_m^\theta)$ and common log dispersion τ. Banks have exact CARA preferences over return on equity, with a common coefficient γ_E and a fixed equity-to-deposit scale $e = 0.10$. The implied coefficient over earnings per deposit is $g = \gamma_E/e$. A common benefit b_V accrues per unit of floating lending. We retain a common lending-conduct coefficient χ_C, with standard Cournot as the special case $\chi_C = 1$, and market-specific constant entry hurdles F_m. There is no rate-dependent risk-aversion coefficient, entry-cost elasticity or partial-entry adjustment parameter.

With endogenous project sizes, converting fixed to floating borrowing does not generally preserve the loan quantity. Bank pricing therefore includes $b_V \, \partial Q_m^V / \partial Q_m^F$, rather than imposing a one-for-one substitution. The derivative is evaluated along borrower demand at a given wage and funding base. In the unit-loan limit it equals -1, recovering the earlier fixed-scale pricing condition. Borrower and bank certainty equivalents are evaluated over the two rate states without a mean–variance approximation.

Rates, production and household saving. The real rates are $r_0 = -0.74653\%$ and $r_a = -1.33653\%$, with a 59 bp gap. These levels correspond approximately to nominal returns of 1.309% and 0.719% less common expected inflation of 2.0555%; the savings return is implied by the selected gap. Operating overhead is included in the entry hurdles. Unlike the fixed-scale model, the level of r_0 now affects project investment and the household saving equilibrium.

We retain the production curvature and log/linear household preferences in Section 4. Table 2 distinguishes fitted financial parameters from real-side assumptions and baseline scale normalisations. Wages, consumption, labour and deposits are solved jointly after each shock. The investment predictions are conditional on the assumed production and labour-supply elasticities, which are not identified by the financial targets alone.

Targets and fit. The calibration matches twelve baseline moments—deposit betas, premiums, HHIs and fixed loan-volume shares in each market—and four response moments: aggregate fixed share rises by 1.3 pp, the premium falls by 3 bp, HHI falls by 0.067, and the high-beta fixed share falls by 5 pp after a 50 bp increase in r_0. The outside return r_a stays fixed. Premium and HHI responses use baseline total loan-volume weights $\omega_m^b = I_m^b / I^b$; the aggregate fixed share is $\sum_m Q_m^F / \sum_m I_m$ at each equilibrium.

Conditional on the real-side assumptions, there are sixteen free financial parameters: five common parameters $(g, \tau, \chi_C, b_V, v_F)$, two independent entrepreneur shares, three borrower distribution locations, three search parameters and three entry hurdles. Baseline HHIs and betas determine counts and search parameters. The other parameters are fitted jointly with both equilibria. All sixteen financial moments are matched, and entrepreneur shares are strictly decreasing. This parameter count alone does not establish identification.

5.2 Responses to a higher financing rate

The experiment raises the expected real financing rate r_0 by 50 bp while holding the outside return r_a, rate uncertainty, preferences, contract benefits and entry hurdles fixed. Banks enter until their certainty equivalents again cover the entry hurdles; households and entrepreneurs simultaneously adjust saving and project sizes. Tables 3 and 4 describe this

Table 2: Parameters and calibration targets

Parameter	Value	Target or calibration basis
Fitted financial parameters		
Log dispersion τ	2.94	Joint fit: Table 3, A–B
Bank risk aversion γ_E	11.53	Joint fit: Table 3, A–B
Lending conduct χ_C	3.13	Joint fit: Table 3, A–B
Bank floating benefit b_V (bp)	12.83	Joint fit: Table 3, A–B
Borrower benefit v_F (bp)	1.41	Joint fit: Table 3, A–B
Borrower medians $e^{\mu_m^\theta}$	(16.05, 8.17, 2.28)	Joint fit: Table 3, A–B
Entrepreneur shares φ_m (%)	(53.16, 26.08, 20.76)	Joint fit: Table 3, A–B
Deposit search ζ_m	(0.09, 0.15, 0.14)	Deposit betas: Table 3, A
Entry hurdles $10^6 F_m$	(553.04, 330.11, 153.59)	HHIs/free entry: Table 3, A
Maintained assumptions		
Equity/deposit ratio e	0.10	Risk-bearing normalisation
Rate uncertainty σ (bp)	50.00	Maintained rate uncertainty
Real financing rate r_0 (%)	-0.75	Nominal benchmark less inflation
Real outside return r_a (%)	-1.34	Implied by the 59 bp rate gap
Rate gap $r_0 - r_a$ (bp)	59.00	Imposed funding–saving gap
Deposit shares δ_m	1/3	Equal deposit relationship shares
Capital exponent α	1/3	Maintained production curvature
Expansion curvature κ	0.10	Maintained investment curvature
Retail elasticity ε	6.00	Maintained retail markup
Discount factor δ_H	0.96	Maintained household preference
Labour exponents (ν_0, ν_1)	(2.00, 2.00)	Maintained labour-supply curvature
Implied quantities and baseline normalisations		
Effective coefficient $g = \gamma_E/e$	115.26	Implied by γ_E and e
Effective bank counts n_m	(4.00, 4.55, 7.69)	Baseline HHIs: $n_m = 1/H_m$
Project productivity A	3.94	Baseline $I^b = L_1^b = 1$
Current productivity A_0	2.05	Baseline $L_0^b = 1$ and goods clearing
Labour scale χ_0	1.62	Baseline $p_0^b = (\varepsilon - 1)/\varepsilon$
Labour scale χ_1	2.19	Baseline $L_1^b = 1$ and labour clearing
Baseline surplus μ^b	0.10	Baseline investment $I^b = 1$

Notes: Market-specific values are ordered by low, middle and high deposit beta. The joint financial fit matches baseline premiums and fixed shares and the four response targets in Table 3; these parameters are estimated together, subject to decreasing entrepreneur shares, not separately identified by individual moments. Table 4 reports the resulting market responses. Borrower distributions are reported by their medians and common log dispersion. Entry hurdles are in profit units. Values are rounded only for presentation.

Table 3: Calibration

Panel A. Baseline targets and model fit

	Low beta	Middle beta	High beta
Deposit beta	0.22	0.35	0.48
Fixed-rate premium (bp)	23.90	14.20	4.30
Lending HHI	0.25	0.22	0.13
Fixed share of loan volume (%)	18.60	18.80	21.50

Panel B. Target responses and real outcomes

Moment	Target	Model
Fixed-loan volume share (pp)	+1.30	+1.30
Fixed-rate premium (bp)	−3.00	−3.00
Lending HHI change (×100)	−6.70	−6.70
High-beta fixed-loan share (pp)	−5.00	−5.00
Investment and deposits (%)	Not targeted	−0.67
Fixed lending (%)	Not targeted	+6.03
Floating lending (%)	Not targeted	−2.27
Current consumption (%)	Not targeted	−0.29
Terminal employment (%)	Not targeted	−0.10

Notes: Panel A reports targets and model values, which coincide at the displayed precision. Panel B reports responses to a 50 bp rate increase after full entry adjustment. Premium and HHI changes use baseline loan-volume weights; fixed share uses actual loan volumes. HHI changes are multiplied by 100. Untargeted real outcomes depend on the real-side assumptions.

fully adjusted equilibrium, rather than the immediate response before entry. Both equilibria pass the entry-stability checks in Appendix C.4.

Table 4: Market responses to a 50 bp rate increase

	Low beta	Middle beta	High beta
Post-shock fixed share (%)	21.63	21.62	16.50
Fixed-share change (pp)	+3.03	+2.82	-5.00
Post-shock premium (bp)	18.96	11.13	6.30
Premium change (bp)	-4.94	-3.07	+2.00
Post-shock HHI	0.17	0.15	0.09
HHI change	-0.08	-0.07	-0.04
Post-shock bank count	5.76	6.66	11.10
Post-shock deposit beta	0.31	0.46	0.58
Fixed-loan quantity change (%)	+15.57	+14.21	-23.88
Total investment change (%)	-0.62	-0.67	-0.82

Notes: Entry adjusts fully. Shares are fractions of loan volume, not borrower counts. Entrepreneur populations remain fixed, while local investment and household deposits adjust.

Entry and the price of protection. At the initial deposit betas, the increase in r_0 raises the expected deposit margin by $(1 - \beta_m^d) \times 50$ bp. This makes entry attractive even though aggregate lending subsequently contracts. Effective bank counts rise from $(4.00, 4.55, 7.69)$ to $(5.76, 6.66, 11.10)$, and the loan-weighted HHI falls by 0.067. The smaller deposit base is shared among more banks. Entry also increases deposit betas from $(0.22, 0.35, 0.48)$ to $(0.31, 0.46, 0.58)$, so funding costs become more responsive to the benchmark.

These changes have opposing effects on fixed-rate pricing. More lenders reduce the premium income a bank sacrifices on its existing loans when it expands fixed credit, encouraging a lower premium. Higher deposit pass-through reduces the earnings exposure that fixed loans hedge, weakening the incentive to supply that protection. The low- and middle-beta premiums fall by 4.94 and 3.07 bp, and fixed shares rise by 3.03 and 2.82 pp. In the high-beta market, the weaker hedging incentive is associated with a 2.00 bp premium increase and a 5.00 pp fixed-share decline.

Lending composition and aggregate investment. The low- and middle-beta markets account for about 79% of baseline lending, so their expansion of fixed borrowing outweighs the high-beta contraction. Aggregate fixed share rises by 1.30 pp, while fixed-loan quantities increase by 15.57% and 14.21% in the first two markets and fall by 23.88% in the third. Overall, fixed lending rises by 6.03% and floating lending falls by 2.27%.

The fixed borrowing rate $r_0 + s_m$ nevertheless rises by 47.00 bp on average: the lower premium only partly offsets the benchmark increase. Higher financing costs reduce project sizes in all markets. Aggregate investment and household deposits fall together by 0.67%,

while current consumption falls by 0.29% and terminal employment by 0.10%. Wage adjustment moderates the contraction. These real outcomes are untargeted implications; the financial responses in Table 3, Panel B, are fitted moments.

5.3 Bank risk-bearing capacity and investment

Greater bank risk-bearing capacity can reduce the supply of fixed-rate protection even when financial preferences, borrower distributions and market structure start from the same calibration. We increase capacity c from 1 to 1.8 by replacing γ_E with γ_E/c in the exact CARA objective:

$$-\mathbb{E}\exp\left(-\frac{\gamma_E}{c}\frac{\Pi_{im}(z)}{E_{im}}\right).$$

The equity-to-deposit normalisation remains $e = 0.10$; c changes effective risk tolerance, not bank funding directly. All other parameters are those in Table 2. We hold bank counts and expected rates fixed to isolate the capacity channel. Household deposits, project sizes, wages and employment continue to adjust in equilibrium. This is an incumbent-bank comparative static; it does not impose free entry again after the capacity change.

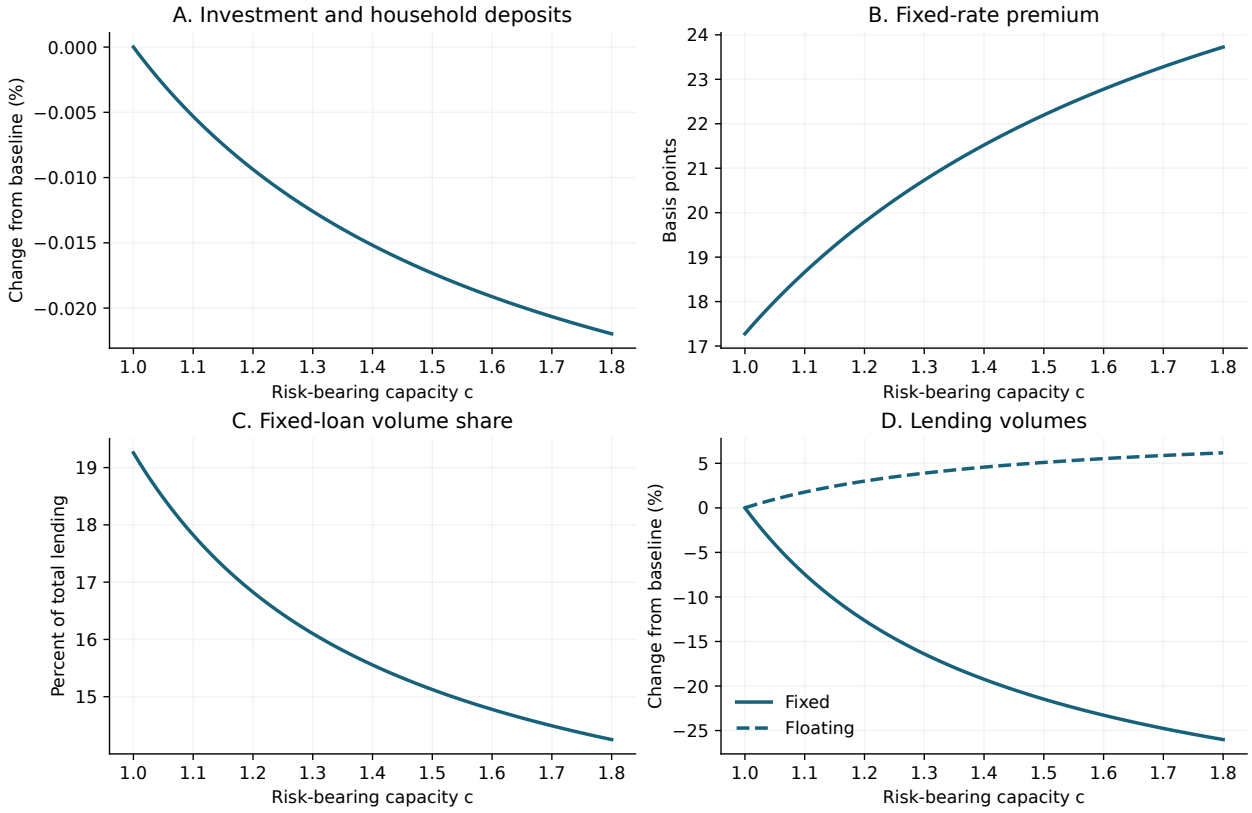


Figure 4: Risk-bearing capacity under the joint CARA calibration

Notes: The baseline is the same as in the monetary experiment. Capacity changes γ_E/c; counts, search parameters, contract benefits, rate levels and borrower preferences remain fixed. Deposits equal endogenous investment. Premiums use baseline total loan-volume weights; fixed shares use actual loan volumes. Quantity changes are relative to $c = 1$.

Figure 4 traces the equilibrium as capacity increases. Increasing capacity by 80% raises the weighted fixed premium by 6.45 bp and reduces the aggregate fixed share by 5.01 pp. Fixed lending falls by 26.01%, while floating lending rises by 6.18%. Total investment and household deposits fall by 0.0220%.

Banks retain positive exposure to the rate shock along this path. Greater risk tolerance lowers the marginal value of hedging, so banks charge more for fixed-rate protection. Borrowers respond both by switching to floating finance and by reducing project size. Substitution between contracts is much larger than the change in total investment. Wage adjustment offsets part of the contraction, but does not reverse it.

Diversification or anticipated public support can motivate changes in effective risk tolerance (Chari & Kehoe, 2016; Froot & Stein, 1998). Here c isolates that channel; it is not a structural estimate of either intervention. The exercise shows that greater bank risk-bearing capacity can reduce borrower protection and investment under the same preferences and baseline calibration used for the monetary responses.

6 Conclusion

This paper shows how relative market power in funding and lending shapes banks' interest rate risk and monetary transmission. Banks with less rate-sensitive funding charge higher fixed–variable loan premiums, even though stable funding makes fixed-rate lending an attractive earnings hedge. Greater lending concentration is associated with higher premiums conditional on funding sensitivity and a weaker expansion of fixed-rate loan stocks after monetary tightening. Our model explains these patterns through banks' tradeoff between earning lending rents and hedging earnings risk. Entry links competition on both sides of the balance sheet, changing both the incentive to hedge and the price of borrower protection.

The joint calibration connects this mechanism to household saving and investment. Following a higher real financing rate, entry lowers aggregate concentration and the fixed-rate premium, while the fixed share of lending rises. Fixed credit expands even as total investment and household deposits contract: cheaper protection partly cushions, but does not offset, higher overall financing costs. The responses differ across markets because stronger lending competition and weaker funding hedges have opposing effects on fixed-rate pricing.

Using the same calibration, greater bank risk-bearing capacity raises premiums and reduces fixed-rate lending when expected rates and bank counts are held fixed. Banks become less willing to supply protection as a hedge, and borrowers switch towards floating finance and reduce project scale. Investment falls as a result. Diversification and anticipated public support may operate through this channel, making lending terms and borrower investment relevant to assessing changes in bank resilience.

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A Empirical Appendix

A.1 SBA 7(a) Loan-Level Summary Statistics

Table 5: Descriptive statistics of SBA 7(a) loans by rate type, 2010Q1–2019Q4

	Fixed-rate				Variable-rate			
	Mean	P25	P50	P75	Mean	P25	P50	P75
Number of banks	1,013				1,643			
Number of loans	33,207				115,741			
Gross approval ($ billions)	27.8				97.2			
Approval amount ($ thousands)	837	294	560	1,071	840	239	492	1,053
Guarantee share (%)	76.7	75.0	75.0	75.0	77.6	75.0	75.0	75.0
Interest rate at origination (%)	5.86	5.25	5.85	6.50	6.08	5.50	6.00	6.65
Maturity (months)	182	116	120	300	172	120	120	300
Jobs supported	15.8	3.0	7.0	18.0	15.5	3.0	8.0	17.0

Notes. SBA 7(a) Guaranty loans approved 2010Q1–2019Q4 by depository banks. Approval amount reported in 2019 dollars. The approval amount is the gross approval; the guarantee share is the SBA-guaranteed amount over the gross approval; the rate is the initial contract rate; maturity is the stated term.

A.2 County-level HHI: branch count and SBA 7(a) loan share

Figure 5 compares county-level concentration measured using banks' shares of local branches and their shares of SBA loan dollars. Both measures exhibit substantial geographic variation. Panel (c) shows a positive and statistically significant association: counties with more concentrated branch ownership also tend to have more concentrated SBA lending. This correspondence supports the relevance of branch-network concentration for the local lending environment.

The distinction between the two measures matters for our pricing analysis. SBA lending shares are equilibrium outcomes that reflect banks' loan pricing and supply decisions together with borrower demand. Concentration calculated from contemporaneous loan shares may therefore be jointly determined with the fixed–variable rate differential. Branch-count concentration captures the distribution of established banking presence without directly incorporating current loan volumes. We accordingly use one-year-lagged branch-count HHI to capture local market structure while reducing this direct source of simultaneity. Its positive association with SBA lending concentration supports its use as a proxy for lending-market concentration, conditional on measured funding sensitivity and the other controls. This interpretation does not require branch networks to be exogenous, but it also does not establish a causal effect of concentration on pricing.

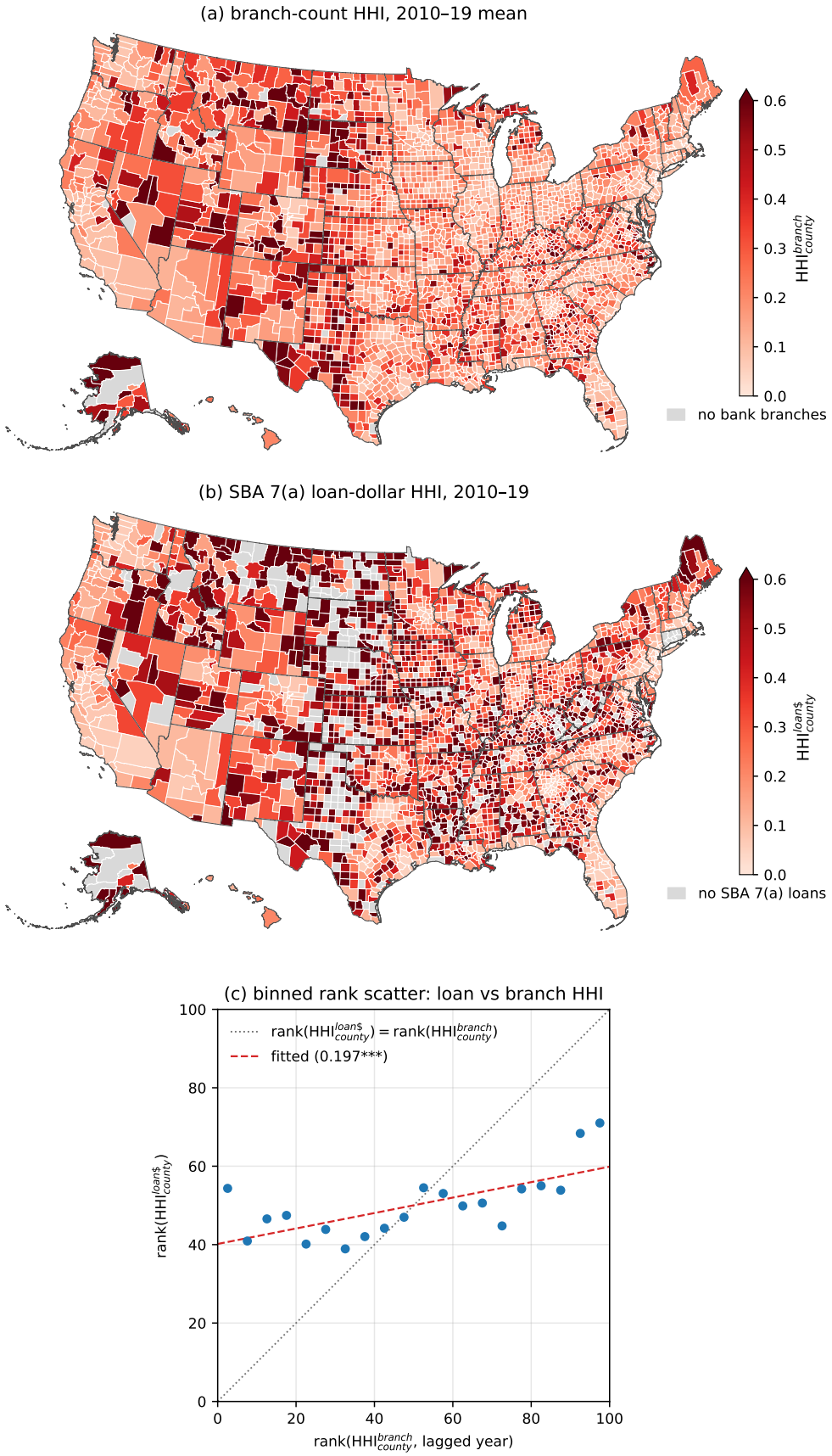


Figure 5: County-level HHI: branch count and SBA 7(a) loan share

B Model Appendix

B.1 Ownership and the bank objective

Bank owners care about the level and variability of their terminal consumption. We assume that a bank's ownership base is proportional to its assets and that owners share earnings equally. Each owner then bears the earnings risk associated with one unit of lending. This gives the bank objective over returns on assets in Section 3.1 and, when entry is endogenous, the participation value in Section 3.6.

Ownership and preferences. Each bank i has a fixed pool $\overline{\mathcal{O}}_i$ of identical prospective owners, large enough to accommodate the bank sizes under consideration. The pools are disjoint, and an owner can participate only through its designated bank. A bank with assets $d_i > 0$ activates a subset $\mathcal{O}_i \subseteq \overline{\mathcal{O}}_i$ of mass d_i. Asset size and membership are fixed before portfolio choices and the interest-rate shock. Prospective owners outside $\mathcal{O}_i$ remain inactive.

Owners have deterministic outside terminal wealth b and mean–variance preferences over terminal consumption,

$$\mathcal{U}(c) = \mathbb{E}c - \frac{\gamma}{2}\operatorname{Var}(c), \qquad \gamma > 0.$$

The coefficient γ is common across owners and independent of bank size. An inactive owner consumes b, giving value $\mathcal{U}(b) = b$. Owners have no other exposure to the interest-rate shock.

Given bank counts. First hold the number of banks fixed and omit participation costs. Let $\Pi_i(z) = d_i\pi_i(z)$ denote total net interest earnings, where $\pi_i(z)$ is earnings per unit of assets. Equal sharing among the mass d_i of active owners gives each owner terminal consumption

$$c_i^0(z) = b + \frac{\Pi_i(z)}{d_i} = b + \pi_i(z).$$

Since outside wealth is deterministic, the owner's certainty-equivalent gain is

$$\mathcal{U}(c_i^0(z)) - \mathcal{U}(b) = \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \equiv W_i. \tag{27}$$

All owners therefore prefer the fixed-rate portfolio share that solves

$$\max_{l_i^F \in [0,1]} \left\{ \mathbb{E}\pi_i(z) - \frac{\gamma}{2}\operatorname{Var}(\pi_i(z)) \right\},$$

taking rival portfolios as given and anticipating the deposit continuation. This is the bank objective in Section 3.1. Mean–variance preferences are specified over owners' consumption, and equal sharing translates them into preferences over the bank's return on assets.

Summing the owners' gains gives

$$d_i W_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)),$$

where $\operatorname{Var}(\Pi_i(z)) = d_i^2 \operatorname{Var}(\pi_i(z))$. Dollar risk tolerance is d_i/γ, proportional to the mass of owners. This ownership interpretation is consistent with Wilson (1968), who shows that a syndicate's risk tolerance equals the sum of its members' risk tolerances under efficient sharing. In the symmetric benchmark, total assets are one and $d_i = 1/N$. Each bank then

has risk tolerance $1/(N\gamma)$, and aggregate risk tolerance across banks is $1/\gamma$. Increasing N divides both assets and active ownership among more banks.

Endogenous entry and participation costs. Now let banks choose whether to operate, with assets assigned according to the active bank counts as in Section 3.6. Ownership follows the same allocation rule, so a participating bank with assets d_i has active owner mass d_i. Participation costs $F > 0$, paid from owners' outside resources in terminal-value units. Active owners share the cost equally, giving each terminal consumption

$$c_i^F(z) = b + \frac{\Pi_i(z) - F}{d_i} = b + \pi_i(z) - \frac{F}{d_i}.$$

The allocation satisfies $\int_{\mathcal{O}_i}(c_i^F(z) - b)\,\mathrm{d}j = \Pi_i(z) - F$. Because the fee is deterministic, the owner's net gain is

$$\mathcal{U}(c_i^F(z)) - \mathcal{U}(b) = W_i - \frac{F}{d_i}.$$

Conditional on participation and bank size, every owner chooses the portfolio that maximises W_i. All active owners also agree on participation, which is weakly preferred when $W_i \geq F/d_i$.

Writing $c_i^F(j, z) = b$ for an inactive owner, the sum of gains across the bank's fixed pool of prospective owners is

$$V_i = \int_{\overline{\mathcal{O}}_i} \left[\mathcal{U}(c_i^F(j, z)) - b\right] \mathrm{d}j = d_i W_i - F.$$

In terms of total earnings, this gives

$$V_i = \mathbb{E}\Pi_i(z) - \frac{\gamma}{2d_i}\operatorname{Var}(\Pi_i(z)) - F. \tag{28}$$

A larger bank draws on more owners with the same individual risk tolerance and spreads the fixed cost over a larger ownership base. The common interest-rate shock affects all owners. For a given per-asset earnings distribution, dollar earnings variance grows with the square of bank size, while the coefficient on that variance falls in proportion to size.

A related entry formulation appears in Atkeson et al. (2015), who model banks as coalitions of risk-averse traders and compare per-trader certainty-equivalent gains with a fixed entry cost divided by coalition size. Their coalition sizes are given. Here active ownership follows the asset allocation implied by bank counts, so both size and ownership adjust with entry.

Symmetric entry. Consider a symmetric configuration with market size A, M local deposit communities, and n banks per community. Each bank has assets and active owner mass $d = A/(Mn)$. In the deposit retention continuation, $\beta^d = 1 - 1/n$. Evaluating (4) at the symmetric portfolio $l_i^F = L^F(n; M)$ gives per-asset earnings

$$\pi(z) = \frac{r_0 - r_a}{n} + L^F(n; M)s(n; M) + \left[\frac{1}{n} - L^F(n; M)\right]\sigma z.$$

Since $\mathbb{E}z = 0$ and $\mathrm{Var}(z) = 1$, substitution into (27) yields

$$W(n; M) = \frac{r_0 - r_a}{n} + L^F(n; M)s(n; M) - \frac{\gamma}{2}\sigma^2 \left[\frac{1}{n} - L^F(n; M)\right]^2.$$

This is the per-asset franchise value in (12). The first term is the expected funding advantage, the second is premium income from fixed-rate lending, and the final term is the cost of residual earnings risk borne by each owner.

Using $d = A/(Mn)$ in (28), the participating bank's net value is

$$V(n; A, M) = \frac{A}{Mn}W(n; M) - F = Av(n; M) - F, \qquad v(n; M) = \frac{W(n; M)}{Mn}.$$

The owners' participation condition is therefore equivalent to $Av(n; M) \geq F$. At fixed bank counts, a larger market expands both bank assets and active ownership, increasing total certainty-equivalent earnings relative to the fixed participation cost. Entry changes bank size and membership through the allocation rule. An additional entrant evaluates its value using its post-entry size and the reoptimised deposit and lending continuation, as in (32) and (33). This appendix verifies deposit pricing, establishes the value inequalities used in the entry analysis, and characterises the continuation following entry into a single community.

B.2 Deposit-pricing continuation

Let $\mu = r_0 - r_a > 0$ and $c = 1/n$. Risk-neutral depositors rank offers $r_a + \alpha(r_1(z) - r_a)$ by their expected return $r_a + \alpha\mu$, so larger α is preferred even if $r_1(z) < r_a$ in one state. For integer counts, fix any feasible loan-portfolio profile. We construct a continuation in which both eligible banks offer $\alpha = 1$ on contested accounts; the incumbent chooses its offer on uncontested accounts below. The resulting symmetric pricing conditions also define the continuous extension in n and N used in the main text.

Normalise funding expenses by the bank's committed loan book. As a reference, paying r_a on uncontested accounts and $r_1(z)$ on contested accounts gives exposure $G_i^0 = c - l_i^F$. Against rivals' contested-account offers of one, any unilateral bid profile changes funding expense relative to this reference by $t(r_1(z) - r_a)$ with $t \geq 0$. Losing a contested account replaces funding at the same wholesale rate; attracting a rival's account requires an offer above that rate. Raising an uncontested offer also increases t. The bank therefore chooses the net funding adjustment to maximise

$$-\mu t - \frac{\gamma\sigma^2}{2}(c - l_i^F - t)^2, \qquad t \geq 0.$$

Its unique optimum and resulting average pass-through are

$$t_i^* = \left[c - l_i^F - \frac{\mu}{\gamma\sigma^2}\right]_+, \qquad \beta_i^d = 1 - c + t_i^*. \tag{29}$$

Since $0 \leq t_i^* < c$, this optimum is attained by offering $\alpha = t_i^*/c < 1$ on uncontested accounts and one on contested accounts. All deposits are retained under the tie rule. The optimisation covers all unilateral bid deviations, including bids above one to obtain additional funding as a hedge. Thus these offers are best responses after every portfolio profile. We do not claim

uniqueness of all deposit-bidding equilibria.

Proof of Lemma 1. Hold the other banks' fixed quantities constant. Since $L^F = (1/N)\sum_j l_j^F$, a unilateral increase in l_i^F lowers the market premium at rate $\partial s/\partial l_i^F = -\theta\sigma^2/N$. Anticipating its optimal deposit offer, bank i chooses l_i^F to maximise

$$V_i = c\mu + l_i^F s(L^F, \sigma) - \mu t_i^* - \frac{\gamma\sigma^2}{2}(c - l_i^F - t_i^*)^2.$$

The envelope theorem gives

$$\frac{\partial V_i}{\partial l_i^F} = s - \frac{\theta\sigma^2 l_i^F}{N} + \min\{\gamma\sigma^2(c - l_i^F), \mu\}.$$

This derivative is continuous and strictly decreasing in the bank's own share. Away from the switching point, the second derivative is $-2\theta\sigma^2/N$ where $t_i^* > 0$ and $-\sigma^2(2\theta/N + \gamma)$ where the adjustment constraint is strictly binding at $t_i^* = 0$, establishing global strict concavity. At a symmetric allocation the first-order condition is

$$\theta\sigma^2\left[1 - (1 + 1/N)L^F\right] + \min\{\gamma\sigma^2(c - L^F), \mu\} = 0. \tag{30}$$

Its left-hand side is strictly decreasing, positive at zero and negative at one, so it has one interior root, including for real competition indices $n, N \geq 1$. In the maintained regime $t_i^* = 0$ at this root, and the minimum in (30) equals $\gamma\sigma^2(c - L^F) = \gamma\sigma^2 G$. Substituting $s = \theta\sigma^2(1 - L^F)$ and rearranging gives (5). Strict own concavity ensures that each bank's choice is a best response to the common choice of its rivals. □

To characterise the solution in both deposit-pricing regimes, define

$$L_B(n, N) = \frac{\theta + \gamma/n}{\theta(1 + 1/N) + \gamma}, \qquad L_D(N) = \frac{N}{N + 1}\left(1 + \frac{\mu}{\theta\sigma^2}\right).$$

The joint solution is

$$L^{F*} = \min\{L_B, L_D\}, \qquad s^* = \theta\sigma^2(1 - L^{F*}),$$
$$\beta^{d*} = 1 - \frac{1}{n} + \left[\frac{1}{n} - L^{F*} - \frac{\mu}{\gamma\sigma^2}\right]_+.$$

Writing $G_B = 1/n - L_B$, voluntary additional pass-through is inactive exactly when $\mu \geq \gamma\sigma^2 G_B$. Then equations (5) and (6) apply. The condition below ensures this regime uniformly over real $n \geq 1$. For $n \geq 2$ it is automatic: the left-hand side of (30) is positive at $L^F = c$, so the root has $L^F > c$ and hence $t_i^* = 0$.

B.3 A sufficient condition for competition-driven deposit pass-through

Fix $M \geq 1$ and let $N = Mn$ as n varies over $[1, \infty)$. A sufficient condition for $\beta^d = 1 - 1/n$ at every symmetric equilibrium along this comparison is

$$\mu = r_0 - r_a \geq \frac{\gamma\sigma^2\theta}{\theta(M+1) + \gamma M}. \tag{31}$$

To verify it, set $D = \theta(1 + 1/N) + \gamma$ and write $G_B = c - (\theta + \gamma c)/D$. Holding the other argument fixed gives

$$\frac{\partial G_B}{\partial c} = \frac{\theta(1+1/N)}{D} > 0, \qquad \frac{\partial G_B}{\partial N} = -\frac{\theta(\theta+\gamma c)}{N^2 D^2} < 0.$$

Along $c = 1/n$ and $N = Mn$, increasing n lowers c and raises N, so G_B is strictly decreasing. Its maximum occurs at $n = 1$, where

$$G_B(1, M) = \frac{\theta}{\theta(M+1) + \gamma M}.$$

Condition (31) therefore implies $\mu \geq \gamma\sigma^2 G_B$ for every $n \geq 1$, giving $L^{F*} = L_B$ and zero additional pass-through. It also permits equality at $n = 1$; the envelope condition remains valid at that switching point. This is an equilibrium restriction, not a claim that additional pass-through is never optimal after a deviation in loan portfolios; those deviations are covered by (29).

Equivalently, the condition requires market breadth to satisfy

$$M \geq \underline{M} \equiv \frac{\theta}{\theta + \gamma}\left(\frac{\gamma\sigma^2}{\mu} - 1\right).$$

The negative premium–beta result in Proposition 2 also requires $M < \overline{M} \equiv \theta^2/[\gamma(\theta+\gamma)]$. Thus sufficient joint restrictions are

$$\max\{1, \underline{M}\} \leq M < \overline{M}.$$

For continuous market breadth this interval is nonempty exactly when $\overline{M} > 1$ and $\mu > \gamma^2\sigma^2/(\theta+\gamma)$; when M is a literal number of markets, the interval must contain an integer. A larger M helps keep voluntary deposit hedging inactive but can reverse the premium slope, so a lower bound on M alone does not establish the negative relationship. If $\mu \geq \gamma\sigma^2$, then $\underline{M} \leq 0$ and no additional lower bound beyond $M \geq 1$ is needed.

For the entry extension only, the additional assumption $\mu > \gamma\sigma^2$ makes $t_i^* = 0$ after every feasible portfolio profile, including when a local market has one bank. The same argument applies with heterogeneous counts using $c_i = 1/n_i$ and each bank's own asset normalisation. This supplies the continuation used in the entry derivations below.

B.4 Expected utility and symmetric Cournot lending

Throughout this comparison the deposit contracts are given. For either utility function in Section 3.4, the marginal borrower's premium solves

$$u_\varrho(c-s) = \tfrac{1}{2}[u_\varrho(c+\sigma) + u_\varrho(c-\sigma)], \qquad \varrho = 2\theta(1-L).$$

Greater risk aversion reduces the certainty equivalent of the variable loan, so $s_L < 0$. Under both preferences $s(0) > 0$ and $s(1) = 0$, using the risk-neutral limit at $L = 1$.

Proof of Lemma 2. Fix rival quantities and let l be the bank's own fixed share, so $dL/dl = 1/N$. Write $\mu = r_0 - r_a$. Terminal wealth in the two states is $w(\pm 1) = E + (1-\beta^d)\mu + ls(L) \pm \sigma(1-\beta^d - l)$. A marginal fixed loan pays $s + (l/N)s_L \mp \sigma$ in those states. Weighting these payoffs by marginal utility, dividing by average marginal utility, and then setting $l = L$ gives (9) with

$$H = \sigma \frac{u'(w(-1)) - u'(w(+1))}{u'(w(-1)) + u'(w(+1))}.$$

Substituting the marginal utilities and using $w(\pm 1) = W \pm \sigma G$ gives $H = \sigma \tanh(\gamma\sigma G)$ under CARA and $H = \sigma \tanh[\gamma \operatorname{arctanh}(\sigma G/W)]$ under CRRA. For CRRA we restrict attention to primitives giving positive wealth at every feasible portfolio. □

Proof of Proposition 3. Write $R(L) = s + (L/N)s_L$. For CARA, inverse demand is strictly decreasing and concave. Indeed, for $v(\varrho) = \log\cosh(\varrho\sigma)/\varrho$ and $x = \varrho\sigma$, $\varrho^3 v''(\varrho)$ equals $x^2/\cosh^2 x - 2[x\tanh x - \log\cosh x]$. It vanishes at zero and has derivative $-2x^2\tanh(x)/\cosh^2 x < 0$. The cutoff $\varrho = 2\theta(1-L)$ is affine, which gives the claimed curvature. Thus own premium revenue $ls(L)$ is strictly concave in l, and R is strictly decreasing. For CRRA these two shape restrictions are imposed. State wealth adds an affine term to own revenue, so increasing concave utility gives strict own concavity in both cases.

Now $R(0) > 0 > R(1)$, hence L_C is unique and interior. The hedging term decreases in L (under CRRA use the restriction on G/W) and has the sign of $L_H - L$. At L_C the sign of $R + H$ is the sign of $L_H - L_C$, while at L_H it is the sign of $L_C - L_H$. The unique root is between these shares, or equals their common value. If $L_H = 1$, the root is strictly below one. Own concavity makes this common portfolio a best response for every bank. □

Proof of Proposition 4. Let $F = s + [(1-\beta^d)/M]Ls_L + H$. Implicit differentiation gives $ds/d\beta^d = -s_L F_{\beta^d}/F_L$. Since $s_L < 0$ and $F_L < 0$, the premium falls exactly when $F_{\beta^d} = L(-s_L)/M + H_{\beta^d} > 0$. At fixed L, the CARA hedging term gives $-H_{\beta^d} = \gamma\sigma^2/\cosh^2(\gamma\sigma G)$. For CRRA put $x = \sigma G/W$. Differentiating both exposure and expected equity gives

$$-H_{\beta^d} = \frac{\gamma\sigma^2(W - \mu G)}{(W^2 - \sigma^2 G^2)\cosh^2\big(\gamma \operatorname{arctanh} x\big)}.$$

Here $W - \mu G = E + L(s+\mu) > 0$. At zero bank risk aversion $H = H_{\beta^d} = 0$, while $L_C(-s_L)/M > 0$. Since $R'(L_C) < 0$, an interior equilibrium and the strict inequality persist locally by continuity and the implicit function theorem. This comparison varies bank risk aversion while holding borrower preferences fixed. □

B.5 Uncertainty and the welfare cost of lending power

Proof of Proposition 5. Write $b = 1-\beta^d$, $h = 1/N$, and $v(a) = \log\cosh(a\sigma)/a$, with $v(0) = 0$. Premium payments cancel in the sum of monetary certainty equivalents. The risk cost at a symmetric fixed share L is therefore

$$R(L) = \frac{1}{2\theta}\int_0^{2\theta(1-L)} v(a)\,da + \frac{1}{\gamma}\log\cosh\big(\gamma\sigma(b-L)\big).$$

The first term sums the risk premia of borrowers on variable rates; the second prices bank earnings risk. Inverse demand is $s = v(2\theta(1-L))$. Write $B(L) = -R'(L) = s+\sigma\tanh(\gamma\sigma(b-L))$. Since $s_L < 0$, B is strictly decreasing. It changes sign between zero and one. The efficient share solves $B(L^\star) = 0$, while the Cournot share solves $B(L^F) + hL^F s_L(L^F) = 0$. Hence $B(L^F) > 0$, $L^F < L^\star$, and

$$0 < \Delta(\sigma) = \int_{L^F}^{L^\star} B(u)\,du \le (L^\star - L^F)hL^F[-s_L(L^F)].$$

For small σ, Taylor expansion of $\log\cosh x$ gives the mean–variance risk cost in (11) with opposite sign, up to $O(\sigma^4)$ uniformly in L. The two first-order conditions divided by σ^2 converge to their mean–variance counterparts, each with a unique root. Thus $\Delta(\sigma)/\sigma^2$ converges to the strictly positive mean–variance loss per unit variance.

For large σ, fix any $0 < \varepsilon < 1-b$. On $[b, b+\varepsilon]$ the marginal borrower's coefficient $a = 2\theta(1-L)$ stays bounded away from zero. The identity $\log\cosh x = x-\log 2+\log(1+e^{-2x})$ gives $s = \sigma - (\log 2)/a + o(1)$ and $s_L = -2\theta(\log 2)/a^2 + o(1)$ uniformly there. At $L = b$ the bank hedging term is zero, so both marginal conditions are positive for large σ. At $L = b+\varepsilon$ the hedging term is $-\sigma + o(1)$, making B negative, and the Cournot term is negative as well. Both unique roots therefore converge to b. Since s_L remains bounded near b, the upper bound above implies $\Delta(\sigma) \to 0$. The two roots vary continuously with $\sigma > 0$ by strict monotonicity of their marginal conditions, so the loss is continuous. Its positivity and zero limits at both ends imply an interior global maximum. The argument requires neither a unique peak nor feasibility of the deposit-pricing continuation for arbitrarily large shocks. □

B.6 Franchise value and the participation comparative statics

Maintain the entry extension's additional assumption $\mu \equiv r_0 - r_a > \gamma\sigma^2$. Let $x = 1-\beta^d$, and let $\widehat{W}(\beta^d, N)$ be symmetric per-asset value when beta and lender counts are treated as separate arguments. The portfolio condition implies

$$\widehat{W} = x\mu - \frac{\gamma}{2}\sigma^2x^2 + \sigma^2\left(\frac{\theta}{N}+\frac{\gamma}{2}\right)(L^F)^2.$$

The first two terms are nonnegative because $\mu > \gamma\sigma^2$ and $x \in [0,1]$, while the last is positive. Using $L^F_{\beta^d} = -\gamma/D$ and $L^F_N = \theta L^F/(N^2 D)$ gives

$$\widehat{W}_{\beta^d} = -\mu + \gamma\sigma^2 x - \frac{2\gamma\sigma^2(\theta/N + \gamma/2)L^F}{D} < 0,$$
$$\widehat{W}_N = -\frac{\theta^2\sigma^2(L^F)^2(1-1/N)}{N^2 D} \leq 0.$$

Along $\beta^d = 1 - 1/n$, $N = Mn$,

$$W_n = \frac{\widehat{W}_{\beta^d}}{n^2} + M\widehat{W}_N < 0, \qquad v_n = \frac{W_n}{Mn} - \frac{W}{Mn^2} < 0.$$

The per-asset value remains bounded as $n \to \infty$, so $v \to 0$. For $AW(1;M)/M > F$, continuity and monotonicity yield a unique real $n > 1$ solving (13). Log differentiation yields

$$\frac{d\log n}{d\log A} = \frac{1}{1 - nW_n/W} \in (0,1).$$

Since $\beta^d = 1 - 1/n$ and $N = Mn$, this elasticity implies (14) and completes the proof of Proposition 6. Entry grows less than proportionally with market size, but more local rivals bid for deposits and more lenders compete for borrowers.

B.7 Entry with integer bank counts

A single local market: exact integer entry. For $M = 1$, all banks reoptimise deposit and lending choices when entry raises n to $n+1$. An active integer equilibrium with $n \geq 1$ requires

$$Av(n;1) \geq F \geq Av(n+1;1). \tag{32}$$

Incumbents cover F and another entrant cannot profit. If $Av(1;1) < F$, the market is inactive. At equality either zero or one bank can operate.

Define $A_m = F/v(m;1)$. By Appendix B.6, these thresholds are strictly increasing and unbounded. A count $m \geq 1$ satisfies (32) exactly when $A \in [A_m, A_{m+1}]$. Thus market growth raises the bank count in steps, with two adjacent counts possible at a threshold. Both deposit beta and the number of lenders are nondecreasing in A, and Proposition 2 gives the direction of the premium change across counts.

Several local markets: unilateral entry. For $M > 1$, (13) approximates symmetric participation. An individual entrant changes only its home community's count. Banks then differ in size and beta, so valuing entry requires reoptimising their portfolios.

Let $\mathbf{n} = (n_1, \ldots, n_M)$ be a positive integer vector of local counts, with $N = \sum_{j=1}^M n_j$, and let $v_j(\mathbf{n})$ denote a bank's risk-adjusted continuation value in community j, per unit of aggregate market size. Appendix B.8 derives this continuation. An active integer entry equilibrium satisfies, for each community,

$$Av_j(\mathbf{n}) \geq F, \qquad Av_j(\mathbf{n} + \mathbf{e}_j) \leq F, \tag{33}$$

where $\mathbf{e}_j$ adds one bank in its home community j. Banks compare participation with staying

out ex ante, anticipating portfolio adjustment. After entry the cost is sunk.

From symmetry the incumbent's value per unit of market size is $v(n; M)$. Entry in the first community gives $(n+1, n, \ldots, n)$, whereas $(n+1, \ldots, n+1)$ requires entry everywhere. Appendix B.9 illustrates both premium slopes under unilateral entry.

B.8 Continuation equilibrium with unequal local counts

Take a vector $\mathbf{n} = (n_1, \ldots, n_M)$ with every $n_j \geq 1$ and $N = \sum_{j=1}^{M} n_j$. A bank in community j has assets $d_j = A/(Mn_j)$, global loan-capacity weight $w_j = d_j/A = 1/(Mn_j)$, and deposit beta $\beta_j^d = 1 - 1/n_j$. Banks in that community have identical primitives. Let l_j^F denote their common fixed share. Aggregate fixed lending as a fraction of total credit is $L^F = M^{-1} \sum_j l_j^F$.

Write $h = s/\sigma^2$. The portfolio condition, including boundary choices, gives

$$l_j^F(h) = \min\left\{1, \max\left\{0, \frac{h + \gamma/n_j}{\gamma + \theta/(Mn_j)}\right\}\right\}. \tag{34}$$

Market clearing requires

$$h = \theta\left[1 - \frac{1}{M}\sum_{j=1}^{M} l_j^F(h)\right].$$

The left side minus the right side is continuous and strictly increasing. It is negative at $h = 0$, since $l_j^F(0) < 1$ for every j, and positive at $h = \theta$. Thus there is a unique root in $(0, \theta)$. Equation (34) gives global best responses by own concavity.

Uniqueness also holds without imposing within-community symmetry. Index all banks by i, with capacity weights $w_i > 0$ summing to one and $x_i = 1 - \beta_i^d$. The lending game has the strictly concave weighted potential

$$\mathcal{P}(\mathbf{l}) = \theta \sum_i w_i l_i - \frac{\theta}{2}\left(\sum_i w_i l_i\right)^2 - \frac{\theta}{2}\sum_i w_i^2 l_i^2 + \gamma \sum_i w_i x_i l_i - \frac{\gamma}{2}\sum_i w_i l_i^2.$$

Its derivative divided by w_i equals the derivative of bank i's objective divided by σ^2. The Hessian is $-\theta \mathbf{w}\mathbf{w}' - \mathrm{diag}(\theta w_i^2 + \gamma w_i)$, which is negative definite. Hence the Nash portfolio allocation is unique and symmetric among banks with identical local primitives.

With the solution $h(\mathbf{n})$ and shares $l_j^F(\mathbf{n})$, define

$$v_j(\mathbf{n}) = \frac{1}{Mn_j}\left\{\frac{r_0 - r_a}{n_j} + \sigma^2 l_j^F h - \frac{\gamma}{2}\sigma^2\left(\frac{1}{n_j} - l_j^F\right)^2\right\}. \tag{35}$$

A bank's total continuation value before the fee is Av_j. Recompute (34)–(35) at $\mathbf{n} + \mathbf{e}_j$ to evaluate entry. The positive-count entry conditions (33) follow. For symmetric configurations, all communities face the same entry test. We analyse active configurations rather than characterise all equilibria with inactive communities.

B.9 Numerical illustration of integer entry

Table 6 reports symmetric configurations satisfying participation and no-profitable-entry conditions, using a constant participation cost F. The entrant's continuation is evaluated

at $(n+1, n, \ldots, n)$, with all bank portfolios reoptimised. The examples establish that both premium-slope regimes can occur with unilateral entry.

Breadth M	Size A	Local banks n	Lenders N	Deposit beta	Premium (bp)
1	2	12	12	0.9167	59.15
1	4	18	18	0.9444	51.92
12	2	2	24	0.5000	32.12
12	4	3	36	0.6667	33.50

Table 6: Market size and integer entry. Parameters are $\theta = 1600$, $\gamma = 160$, $\sigma = 0.005$, $r_0 = 0.04$, $r_a = 0.01$, and $F = 0.001$. Each row satisfies participation and no-profitable-entry conditions after recomputing all loan portfolios following entry into one community. These are illustrative equilibria, not a calibration.

For all four rows in Table 6, evaluating these continuation values gives strictly positive incumbent net values and strictly negative entrant net values. These examples illustrate the entry conditions; they do not establish a general existence or uniqueness theorem for the multi-community entry game.

C Investment Model: Derivations and Checks

The quantitative exercises use one joint equilibrium with exact CARA financial preferences, endogenous project sizes and household saving. This appendix records the demand, pricing and market-clearing conditions underlying both experiments.

C.1 Investment and pricing under CARA

At the committed terminal wage, labour optimisation gives $e = \ell(w_1)k$ and net project return $a(w_1)$ in (21). Write $\mu = a(w_1) - r_0$. For borrower risk aversion t, optimised certainty equivalents are

$$U_F(s) = \max_{k\geq 0}\{(\mu - s + v_F)k - \kappa k^2/2\}, \qquad U_V(t) = \max_{k\geq 0}\{\mu k - \kappa k^2/2 - t^{-1}\log\cosh(t\sigma k)\}.$$

The fixed-contract benefit is nonpecuniary and measured in units of profit. It affects contract and scale choices but is not a cash receipt in the goods accounts. On the interior branch,

$$k_F = \frac{\mu - s + v_F}{\kappa}, \qquad \kappa k_V(t) + \sigma\tanh(t\sigma k_V(t)) = \mu.$$

The floating objective is strictly concave. Its optimised value decreases with t, so $U_V(t^*) = U_F(s)$ defines the cutoff separating floating from fixed borrowers.

Write $z = (\log t^* - \mu_m^\theta)/\tau$, $f = 1 - \Phi(z)$ and $R(t,k) = t^{-1}\log\cosh(t\sigma k)$. Along the cutoff curve,

$$k_F(z) = \sqrt{2U_V(t^*)/\kappa}, \qquad s(z) = \mu + v_F - \kappa k_F(z).$$

The envelope theorem yields

$$s_z = \frac{\tau}{k_F}\{\sigma k_V(t^*)\tanh(t^*\sigma k_V(t^*)) - R(t^*, k_V(t^*))\} > 0.$$

Per entrepreneur, fixed and floating quantities and their cutoff derivatives are

$$q(z) = fk_F, \qquad q_z = -\phi(z)k_F - fs_z/\kappa,$$
$$v(z) = \int_{-\infty}^{z} k_V(e^{\mu_m^\theta + \tau u})\phi(u)\,du, \qquad v_z = k_V(t^*)\phi(z).$$

Thus $Q_m^F = \varphi_m q$, $Q_m^V = \varphi_m v$, $s_{m,Q} = s_z/(\varphi_m q_z)$ and $Q_{m,Q}^V = v_z/q_z$. At the indifferent type $k_V(t^*) < k_F$, so $-1 < Q_{m,Q}^V < 0$: some reduction in fixed credit is a reduction in project scale, rather than a one-for-one switch to floating finance.

Bank certainty equivalents and conduct. For a symmetric bank let $d_m = \mathcal{D}_m/n_m$, $G_m = 1 - \beta_m^d - Q_m^F/\mathcal{D}_m$ and

$$\bar{\pi}_m = (1 - \beta_m^d)(r_0 - r_a) + \frac{Q_m^F s_m + b_V Q_m^V}{\mathcal{D}_m}.$$

Operating earnings per deposit are $\bar{\pi}_m + \sigma G_m z$. CARA over return on equity with coefficient γ_E and $E_m = ed_m$ therefore gives

$$W_m = \bar{\pi}_m - \frac{1}{g}\log\cosh(g\sigma G_m), \qquad g = \gamma_E/e.$$

The bank chooses its fixed quantity, taking the wage and funding base as given. Its perceived premium-income margin is $s_m + \chi_C(Q_m^F/n_m)s_{m,Q}$. The lost floating-service income contributes $b_V Q_{m,Q}^V$, and the marginal hedge contributes $\sigma\tanh(g\sigma G_m)$. Their sum gives the pricing condition in Section 4.3. The conduct coefficient is maintained as a reduced-form specification; numerical equilibrium and entry checks do not establish a Nash implementation for $\chi_C \neq 1$.

C.2 Household saving and implementation

Household utility is log in current consumption and linear in terminal consumption, as in (16). Deposit relationship weights δ_m are independent of borrower populations. Expected gross deposit return is

$$\mathbb{E}R^d = 1 + \sum_m \delta_m\{(1-\beta_m^d)r_a + \beta_m^d r_0\}.$$

The household conditions are

$$C_0^{-1} = \delta_H \mathbb{E}R^d, \qquad \chi_0 L_0^{\nu_0} = w_0/C_0, \qquad \chi_1 L_1^{\nu_1} = w_1.$$

These conditions are imposed in both the monetary and capacity experiments. The terminal wage is determined before the rate fixing but is not held at its baseline across counterfactuals.

The benchmark is conditional on an exogenous real interbank rate. A government facility can implement it by intermediating matching surplus and deficit bank positions at the same state-contingent floating payoff. Since bank balance sheets and current goods-market clearing imply $\mathcal{D} = I$, the facility's net position and statewise tax requirement are zero. This implementation relies on inherited current retail supply contracts and is not a claim that an arbitrary real rate clears a fully flexible-price economy.

Retail prices and baseline normalisations. The final-good aggregator has elasticity ε across retail varieties. Flexible terminal pricing implies intermediate-good price $\bar{p} = (\varepsilon-1)/\varepsilon$, retail output $Y_1 = A\ell^{1-\alpha}I$, and household-owned retail dividends Y_1/ε. Current retail supply prices are set before the surprise. At the initial allocation, the optimal markup gives $p_0^b = \bar{p}$; after the surprise, $p_0 = w_0/A_0$ adjusts while the inherited retail contract remains in force.

We normalise $I^b = L_0^b = L_1^b = 1$, implying $\ell^b = 1$. Given fitted baseline project surplus μ^b, the productivity and preference scales are

$$A = \frac{1 + r_0^b + \mu^b}{\alpha\bar{p}}, \quad \chi_1 = \frac{1-\alpha}{\alpha}(1 + r_0^b + \mu^b), \quad A_0 = C_0^b + 1, \quad \chi_0 = \bar{p}A_0/C_0^b.$$

They are fixed in all counterfactuals. These scale choices do not fix subsequent deposits, investment, wages or employment.

Deposit search and counts. The deposit-search rule is $\beta_m^d = \zeta_m(n_m-1)/[1+\zeta_m(n_m-1)]$. Consequently observed baseline betas and effective counts $n_m^b = 1/H_m^b$ imply $\zeta_m = \beta_m^{d,b}/[(1-\beta_m^{d,b})(n_m^b-1)]$. Counts are continuous competition indices. The calibration's positive gap $r_0 - r_a > \sigma$ makes deposits dominate the outside return state by state when $\beta_m^d > 0$.

C.3 Equilibrium and resource accounts

Terminal labour supply and labour demand imply

$$\ell = I^{-\nu_1/(\nu_1+\alpha)}, \qquad 1 + r_0 + \mu = (1 + r_0^b + \mu^b)I^{-p}, \qquad p = \frac{\nu_1(1-\alpha)}{\nu_1+\alpha}.$$

Conditional on counts, the real equilibrium can therefore be expressed as

$$\log \frac{1 + r_0 + \mu}{1 + r_0^b + \mu^b} + p \log \left[\sum_m \varphi_m \{q_m + v_m\} \right] = 0. \tag{36}$$

This condition and the three pricing equations determine the project surplus and borrower cutoffs. Free entry adds three equations $\mathcal{D}_m W_m/n_m = F_m$ for the monetary experiment. The capacity experiment instead fixes counts at their baseline and replaces g by g/c. Household deposits always satisfy $\mathcal{D} = I$.

Let $J_2 = \sum_m \varphi_m \mathbb{E}k_m^2$, $Q^V = \sum_m Q_m^V$ and $K_B = \sum_m n_m F_m$. Aggregate bank-owner net payouts after overhead, entrepreneur income and household consumption are

$$\begin{aligned} C_1^B(z) &= \sum_m \mathcal{D}_m\{\bar{\pi}_m + \sigma G_m z\} - K_B, \\ C_1^E(z) &= \mu I - \frac{\kappa}{2} J_2 - \sum_m s_m Q_m^F - \sigma z Q^V, \\ C_1(z) &= w_1 \ell I + R^d(z) I + Y_1/\varepsilon. \end{aligned}$$

The common bank benefit is accounted for as net ancillary-service output $b_V Q^V$ accruing to bank owners. The borrower convenience benefit is nonpecuniary. Summing the cash budgets yields, in each rate state,

$$C_1(z) + C_1^E(z) + C_1^B(z) + \frac{\kappa}{2} J_2 + K_B = Y_1 + b_V Q^V.$$

Financial-owner payouts are net income: fixed outside wealth can cover adverse-state losses and cancels from the resource identity. Current resources satisfy $C_0 + I = A_0 L_0$. Local interbank positions $\mathcal{D}_m - I_m$ sum to zero. Thus changes in deposits are endogenous equilibrium changes in household saving, rather than an exogenous funding restriction.

C.4 Numerical checks

Both equilibria match their financial targets, and entrepreneur shares decrease across markets. Direct certainty-equivalent calculations, higher-order integration of borrower demand and finite-difference demand derivatives reproduce the reported solution. Household optimality, bank balance sheets and statewise goods accounts are satisfied.

Entry is locally stable when counts respond to proportional net entry values. All eigenvalues of the coupled entry Jacobian have negative real parts at both equilibria. Wider count scans give positive net entry values below the equilibrium count and negative values above it when the other counts are fixed at equilibrium. All 54 tested starting count configurations converge under simultaneous entry adjustment, with deposits, wages and loan quantities re-solved along each path. These are numerical stability checks, not a proof of global uniqueness. The detailed numerical records accompany the calibration code; they are not additional calibration targets.